



GE VERNOVA

ADMS

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Grid Tabular Builder User Guide

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Contents

1. Introduction	5
2. Displays Overview	6
2.1. Toolbars and Menus	6
2.2. Preview Tab	7
2.3. Display Tab	7
2.3.1. General	8
2.3.2. Title Query	10
2.3.3. Content Query	11
2.4. Columns Tab	12
2.4.1. Arranging Columns	13
2.4.2. Editing Columns	14
2.4.3. Column Editor Window	14
2.4.3.1. Field Options	15
2.4.3.2. Type Options	17
2.4.3.3. Layout Options	21
2.4.3.4. Client Role Visibility Options	22
2.5. Conditional Tests Tab	22
2.6. Logical Conditions Tab	24
2.7. Multiple Items Tab	26
2.8. Context Menus Tab	29
2.8.1. Context Menu Tab Items	31
2.9. UI Properties Tab	33
3. Working with the Grid Tabular Builder	37
3.1. Starting the Grid Tabular Builder	37
3.2. File Locations and Conventions	37
3.3. Integrating Grid Tabular Displays	38
3.3.1. DNOM Mapped Fields, Queries, and Query Helpers	38
3.3.2. Existing Displays	38
3.4. Creating a Tabular Display	38
3.4.1. Example 1: Station/Area List Display	39
3.4.2. Example 2: Station/Abnormal List Display	43
3.4.3. Example 3: Multi-Pane Display	47
3.4.4. Example 4: Multi-Select Display	48
3.4.5. Example 5: Auto-Refreshable Display	51
3.4.6. Example 6: Transformers Tree View Display	51
3.4.7. Example 7: Display with a Row Index	54
4. Standard Grid Tabular Displays	58
4.1. Station Exception	58
4.2. De-Energized Customers	59
4.3. Station Abnormal Summary	59

4.4. Measurement Data	59
4.5. Station Output Summary	60
4.6. DPF Station Results	60
4.7. Feeder Losses and Power Quality Indices	60
4.8. Protection Validation	61
4.9. Mobile Substation List	61
4.10. Tag Summary	61
4.11. Annotation Summary	61
4.12. Abnormal Summary	61
4.13. Base Application Status	62
4.14. Advanced Application Status	62
4.15. Application Input Summary	62
4.16. BLA Station Summary	62
4.17. LVM Station Summary	63
4.18. Stations Equipment Commissioning and Decommissioning	63
4.19. Pending Switches	63
4.20. Find Results Summary	63
5. Using Coded Translation Items	64
6. Action URLs	68
6.1. Navigation URLs	68
6.2. Grid Tabular URLs	70
6.3. Pop-Up URLs	71
6.3.1. DMS Pop-Up URLs	71
6.3.2. OMS Pop-Up URLs	72
6.4. Message Box URLs	73
6.5. DNAF Manual Entry Display URLs	75
6.6. SCADA Display URLs	76
6.7. Attribute Panel URLs	77
6.8. Control URLs	78

1. Introduction

Tabular displays are used to show summary and detailed information about the state of the network - information that is not practical or efficient to display on the geographic diagrams.

Tabular displays can display real-time mode, study mode, and simulator mode data. The main displays are called up from the RTDMS, STDMS, or SIMDMS menus. The values in the displays are often linked to other tabular displays to provide greater detail. Many displays also include links to geographic views, to locate the relevant device.

Tabular displays provide powerful sorting and filtering functions to allow easy access to the required information.

Station Exception							
Station	Area	Locate	De- energized Customers	Abnormal Topology Conditions	High Voltage Violation	Highest Load or Bus Voltage (%)	Low Voltage Violation
ROYAL TROON	WSTC		359	8	No	103.09	No
WINGED FOOT	WSTC		0	0	No	103.40	No
COHO	WSTA		1674	1	No	108.94	Yes
DURBAN	STHD		51	4	No	103.99	No
MARS	STHD		0	0	No	104.49	Yes
SATURN	STHD		0	0	No	103.34	No
STEVENS	STHD		0	0	No	101.74	No
SPYGLASS	STHC		0	0	No	104.91	No
CURLEW	STHB		0	0	No	103.76	No
OAKMONT	STHB		0	0	No	103.64	No
PINEHURST	STHB		0	0	No	105.28	Yes
SUN VALLEY	STHB		0	0	No	103.66	No
PINE VALLEY	STHA		0	0	No	103.29	No
PLUTO	STHA		0	0	No	102.83	No
EQSI	STH		0	0	No	98.43	No
BACHELOR	NEAD		0	0	No	103.62	No
BETHPAGE	NEAC		0	0	No	103.57	Yes
MUIRFIELD	NEAC		0	0	No	103.69	No
NEPTUNE	NEAB		0	0	Yes	124.74	No

Figure 1. ADMS Tabular Display

Grid tabular displays are defined by XML files. The ADMS client reads these files and presents the tabular displays in accordance with all the defined parameters. The tool used to manipulate the display files is the Grid Tabular Builder.

Grid tabular displays are shown by the Grid Tabular Viewer function within the ADMS client. This viewer function automatically includes the Display Filter, Reset Sorting Order, Export to CSV, and Print buttons. These features of the Grid Tabular displays are fixed and cannot be controlled by the Grid Tabular Builder definition.

2. Displays Overview

This chapter describes the controls and tabbed displays of the Grid Tabular Builder.

2.1. Toolbars and Menus

The basic operation of the Grid Tabular Builder is controlled using the menus, toolbar buttons, and a series of tabs at the top of the window.

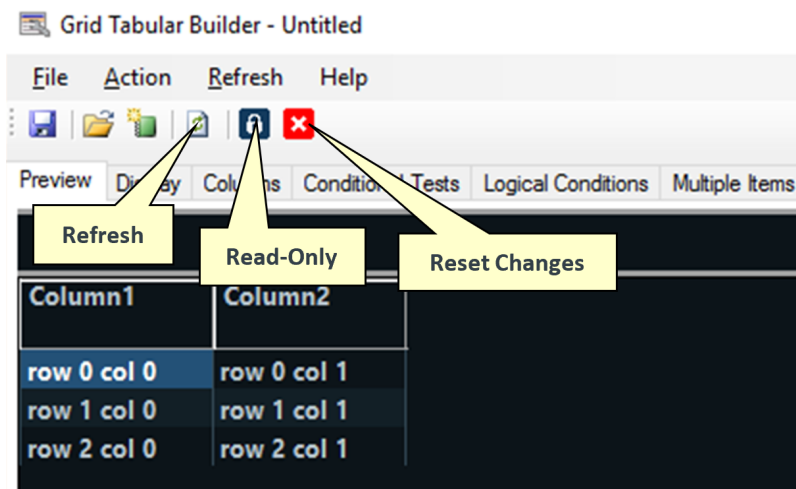


Figure 2. Grid Tabular Builder Controls

In the Grid Tabular Builder, the menus and the toolbar include the following commands:

- **New:** Creates a new, empty grid tabular display file.
- **Open:** Opens a grid tabular display file.
- **Save:** Saves changes to the currently open grid tabular display file.
- **Save As:** Saves all changes to another grid tabular display file.
- **Exit:** Closes the application.
- **Read-Only:** Toggles the read-only mode.

Read-only mode allows the contents of the display to be reviewed, but changes cannot be made. The read-only mode is indicated in the title bar of the Grid Tabular Builder.

- **Reset Changes:** To undo changes that have been made to the display file since the last save, select Reset Change from the Action menu or click the Reset Change button on the toolbar.

Note

The Undo function does not progressively undo incremental changes. It returns the display file back to the point at which the last save was made.

- **Refresh:** Updates the display on the Preview tab to reflect configuration changes.

2.2. Preview Tab

The Preview tab shows how the display will look to the user. It shows the header text, columns, and context menus that have been defined. The Preview tab allows the layout of the display to be easily checked as the display is being built.

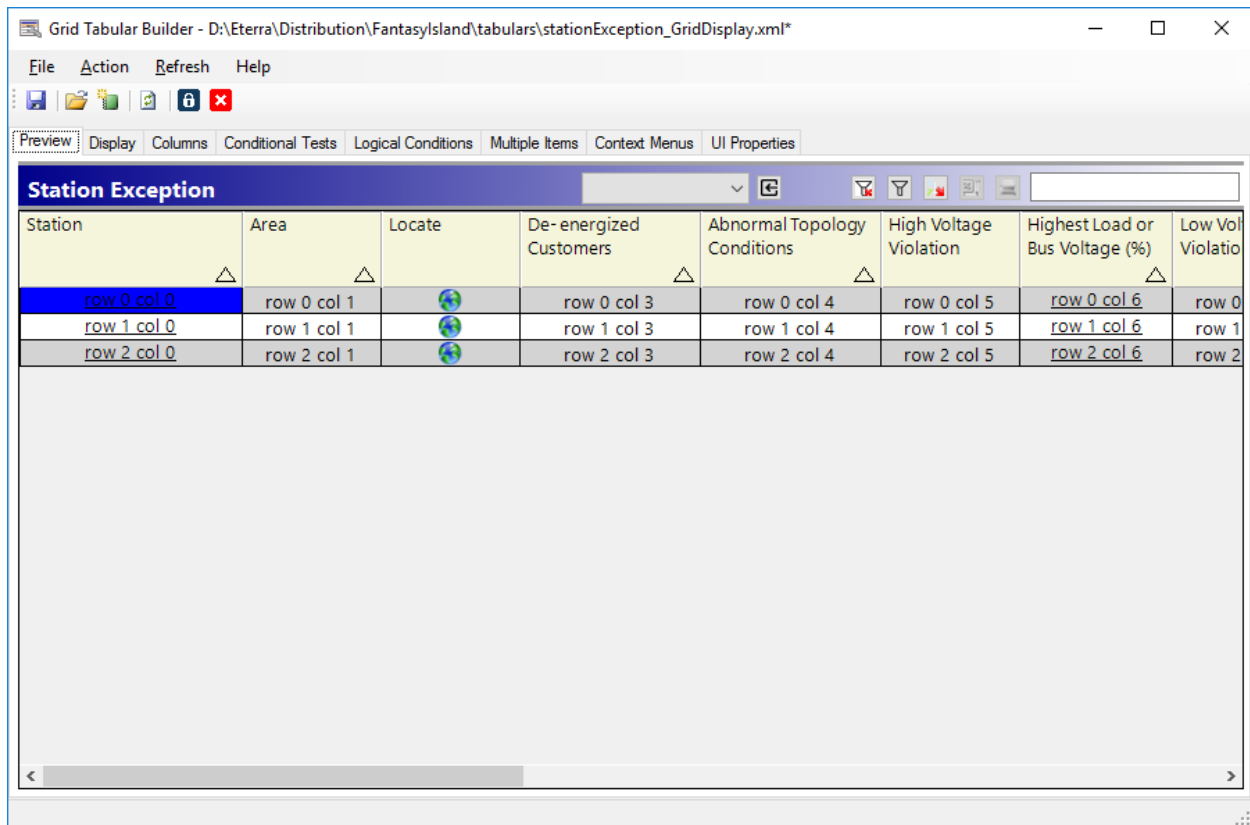


Figure 3. Preview of the Station Exception Display

The Preview display allows the sizes of the columns and rows to be adjusted by dragging the borders. These changes do not affect the values in the display file.

To return the Preview display to the values set in the display file, click the Refresh button on the toolbar or select the Refresh option from the File menu.

2.3. Display Tab

General The Display tab is used to define the parameters that relate to the overall display. The format of each display presents the data in columns across the display, and each row repeats this data for a different display item.

Grid Tabular Builder - D:\Eterra\Distribution\FantasyIsland\tabulars\stationException_GridDisplay.xml

File Action Refresh Help

Preview Display Columns Conditional Tests Logical Conditions Multiple Items Context Menus UI Properties

General

Name: stationException

Title: Station Exception

Empty Tabular Title:

Context Menu:

Initial Sort: DEENER:DESC|ABNORMAL:DESC|STNEXC

Totals Row: None

Refresh Rate: 0

Refresh Duration: 0

Is DMS Query: ☒ Is Local: ☐

Unique ID:

Title Query

From:

Select:

Where:

Content Query

From: %2.SUB

Select: TOPOSUBNAME,LOADTFNAMEFIRST,EQUIPFROMEMS,TFIDFROMSWITCH,NAME,AREANAME,OBJGEOMETRYNAME,STATIONNAME,LOCATEO
BJID,NUMDEENG,NUMABTOT,SUBHIGHVOLT,VIOL,SUBHIGHVOLTID,SUBLOWVOLT,VIOL,SUBLOWVOLTID,SUBV
LTIMBAL,SUBFLWIMBAL,SUBOVERLOADLEVEL,SUBOVERLOAD,SUBGROUNDRETURN,DPFSTATUS,PRVSTATUS,STNEXCPTVHISORT,STNEX
CPTVLOSORT,STNEXCPTVIMBSORT,STNEXCPTFLOWSORT,STNEXCPTGRNDRTSORT,STNEXCPTFLWIMBSORT,SUBCAPAMARGINVIO

Where: PERMITCHECK=1

Figure 4. Display Tab

2.3.1. General

General

Name: stationException

Title: Station Exception

Empty Tabular Title:

Context Menu:

Initial Sort: DEENER:DESC|ABNORMAL:DESC|STNEXC

Totals Row: None

Refresh Rate: 0

Refresh Duration: 0

Is DMS Query: ☒ Is Local: ☐

Unique ID:

The General group includes the following fields:

- **Name:** The name that is used to create the display file. The application adds the "_GridDisplay.xml" suffix to this name (for example, stationException_GridDisplay.xml).

Note

To be recognized by the ADMS client, all display file names must end with

"GridDisplay.xml".

- **Title:** The title of the grid tabular display, as shown in the title bar of the display.
- **Context Menu:** The name of the context menu to be applied to all cells of a grid tabular display. The context menus are defined on the Context Menus tab.
- **Initial Sort:** Defines the priority of the columns in the sorting of the table when it is initially displayed. Columns are identified by their Header Cell Name. For columns to be included in the Initial Sort, a Header Cell Name must be configured on the Columns tab.

Each column is shown with an attribute describing whether it is to be sorted in ascending (ASC) or descending (DESC) order. Each column value is separated with the bar character ("|"). For example:

DEENER:DESC|STATION:ASC

means that the table rows are sorted as follows:

1. The De Energized column (DEENER) in descending order.
2. The Substation column (STATION) in ascending order.

Station Exception					
Station	Area	Locate	De- energized Customers	Abnormal Topology Conditions	High Voltage Violation
<u>3LIONS</u>	WST		425	1	No
<u>WINGED FOOT</u>	WST		20	1	No
<u>RAVEN</u>	WST		1	1	No
<u>BACHELOR</u>	NEA		0	0	Yes

- **Totals Row:** Specifies the appearance of the totals row.
- **Refresh Rate:** The time in seconds between automatic updates of the display. To disable automatic refresh, set this value to 0.
- **Refresh Duration:** The time in seconds that the server keeps the query in memory. This is used to speed up execution of queries for displays that are regularly refreshed. For example, a display may have a refresh rate of 5 seconds and a refresh duration of 60 seconds.

Note

The Refresh Rate and Refresh Duration parameters do not affect OMS Display Plugin grid displays. Some OMS Display Plugin grid displays refresh automatically when data changes. This is a special feature that is not available for standard grid tabular displays.

- **Is DMS Query:** Select this option to direct the data queries to the Distribution Network Object Model (DNOM) database.
- **Is Local:** When this option is selected, the queries are processed locally on the client side. When

not selected, the queries are processed on the server side.

- **Unique ID:** The name of the cell that provides the unique data value for each row of data. This is the argument that is passed to other functions such as actions. For example, in the case of the incident list, the unique ID is ID, the internal identifier for the incident.

2.3.2. Title Query



Title Query

From: %2.SUB(%5)

Select: PFSTATIONEXETIME_TICKS,PFSTATIONTIME_TICKS

Where:

The Title Query fields define the data query that is used to extract the information that appears in the title bar of the display. Each of the values selected by the query is placed directly in the title bar.

For example, the DPF Station Results display (subDPF_GridDisplay.xml) shows the Distribution Power Flow (DPF) evaluation time in the title bar (see following figure), by using this query.

- **Select:** PFSTATIONEXETIME_TICKS,PFSTATIONTIME_TICKS

This example lists data fields to be extracted and displayed in the title bar.

- **From:** %2.SUB(%5)

This example sends the argument to the query helper to extract data from DNOM. The data query is in the Substation Data (.SUB) class. The argument numbers relate to the URL syntax used to call up the display. In the case of this example, the call-up URL for station 3LIONS is:

Netview://RT/gridtabular/subDPF_GridDisplay/3LIONS

Argument 2
(%2)

Argument 5
(%5)

- **Where:** Defines any additional filter conditions that can be applied to the data query.

Note

To use values with spaces in the Where clause, enclose the values in quotes (").

For more information about the DNOM data and query structure, refer to the *ADMS Modeling Reference Guide*.

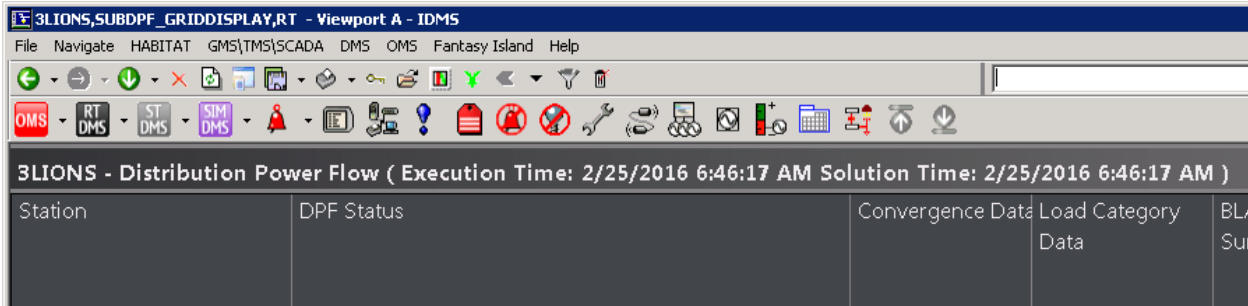


Figure 5. Example of Title Bar Query Display

2.3.3. Content Query

Content Query

From: %2.SUB(%5).ABNRML

Select: ABNDATETIME_TICKS.NAME, LIMITCOMPTYPE, DNOMTABNAME, TOPOSUBNAME.ID, OBJGEOMETRYNAME, STATIONNAME, LOCATEOBJID, FEEDERNAME, TOPOEMSSUBNAME, DNOMRECORDNAME, ABNORMALSTATUS, EQUIPFROMEMS, TFIDFROMSWITCH, COMPHASTAG

Where: TNABNORMAL=ABNORMAL, FEEDERNAME=%6, PERIOD=30

These three fields define the query that is used to extract the information that appears in the data rows of the display.

- **Select:** Defines the data fields that are to be repeated for each row of the display. Values are separated by commas.
- **From:** Defines the arguments sent to the query helper for the extraction of the data from the DNOM (ADMS database). This is expressed as parameter.class (for example, %2.SUB for the SUBSTATION class). The argument numbers relate to the syntax of the display call-up URL (see [Title Query](#)).
- **Where:** Defines any additional filter conditions that can be applied to the data query. "Where" is an optional part of the query and consists of one or more conditions separated by a comma. Multiple conditions are interpreted using "AND" logic. The query result contains only the data that meets all conditions.

A "Where" condition is expressed in the following format:

```
FIELDNAME1 OPERATOR1 VALUE1, FIELDNAME2 OPERATOR2 VALUE2
```

Where *OPERATOR* can contain the following symbols: "=", "!", "<", ">".

Note

To use values with spaces in the Where clause, enclose the values in quotes ("").

Filtering can use either a Query Helper or a Field Map name. For example, the Area Stations display applies "AREANAME = %5" in the Where statement to select only those stations that are

within the required Area of Responsibility. Another common "Where" clause is "PERMITCHECK = 1" to only select data for which the user has permissions.

SQL functions (for example, COUNT) are not supported in "Where" clauses.

i Note

By default, the maximum number of rows that are displayed in a grid tabular display is 10,000 (configured in the `NetServerMainConfig.xml`). To override this value for a particular tabular display, add `MAX_RESULT_ROW = X` to the Where clause, where X is the desired number of rows. To disable the maximum row count limit, set `MAX_RESULT_ROW = -1`.

For more information about the DNOM data and query structure, refer to the *ADMS Modeling Reference Guide*.

2.4. Columns Tab

The Columns tab of the Grid Tabular Builder defines the parameters for each column of the display. This is the core of the display definition. The other tabs of the Builder are used to modify or enhance the parameters defined for each column.

The columns tab displays the read-only settings summary for each column. To change any of the values, the editor must be opened for the required column.

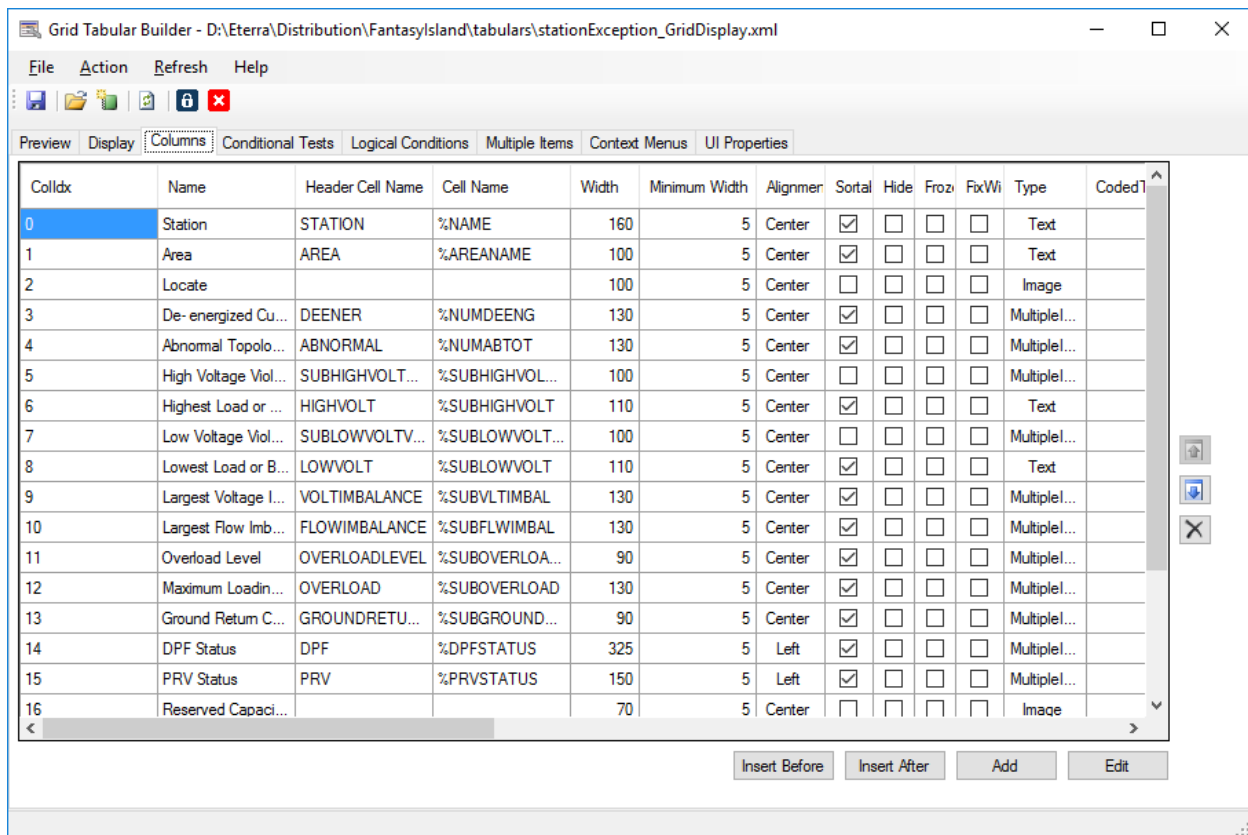


Figure 6. Columns Tab

2.4.1. Arranging Columns

Ordering columns, or inserting and deleting a column, is performed using the buttons at the right and at the bottom-right of the page. Note that the buttons are enabled only when read-only mode for the display file is turned off.

- **Up:** Moves the selected column up one position (i.e., one position to the left on the actual display).
- **Down:** Moves the selected column down one position (i.e., one position to the right on the actual display).
- **Insert Before:** Displays the Add New Column pop-up for a new, blank column that is inserted immediately above the selected column (i.e., immediately to the left on the actual display).
- **Insert After:** Displays the Add New Column pop-up for a new, blank column that is inserted immediately below the selected column (i.e., immediately to the right on the actual display).
- **Add:** Displays the Add New Column pop-up for a new, blank column that is inserted at the bottom of the column list (i.e., at the far right of the actual display).
- **Edit:** Displays the Column Editor pop-up for the selected column.
- **Delete:** Deletes the currently selected column from the display.

2.4.2. Editing Columns

To edit a column:

1. Select the desired column and click the Edit button (or double-click the desired column).

This opens the Column Editor pop-up.

2. Add or change any data in the pop-up, and then click the Update button.

2.4.3. Column Editor Window

The Column Editor window (which shows Add New Column or Edit existing column in the title bar) is used to define all the parameters relating to a single column of the tabular display.

The screenshot shows the 'Edit existing column' window with the following fields and values:

- Field Section:**
 - Name: Largest Voltage Imbalance (%)
 - Header Cell Name: VOLTIMBALANCE
 - Cell Name: %SUBVLTIMBAL
 - ToolTip: Largest Load Voltage Imbalance (%)
 - Action Left: (empty)
 - Action Enter: (empty)
 - Context Menu: (empty)
 - Visibility: None
 - Logical Col Visibility: (empty)
 - Total Type: None
 - Regex: (empty)
- Type Section:**
 - Placement Type: MultipleItems
 - Precision Value: (empty)
 - Alignment: Center
 - Default Width: 130
 - Minimum Width: 5
 - Fix: ☐
 - Hide: ☐
 - Sortable: ☒
 - Frozen: ☐
- Visibility for Client Role:**
 - Excluded Roles: (empty)
 - Included Roles: (empty)
- Buttons:** Update, Cancel

Figure 7. Column Editor Window

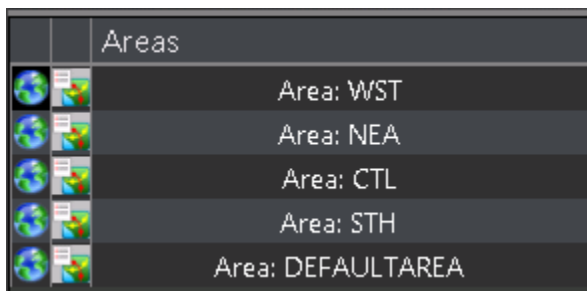
The fields on the Column Editor window are described below.

2.4.3.1. Field Options

The Field group includes the following fields:

- **Name:** The text that appears in the Header Cell for this column.
- **Header Cell Name:** The reference name for this column. The Header Cell Name is used to refer to the column on the Conditional Tests tab. If the Header Cell Name is populated by a DNOM variable, enter the variable name (%name), and the Grid Tabular Builder will automatically update the Title Query Select field on the Display tab.
- **Cell Name:** The name of the value displayed in the data cell. This field can include fixed text as well as variable names.

For example, "Area: %NAME" is displayed as:



	Areas
	Area: WST
	Area: NEA
	Area: CTL
	Area: STH
	Area: DEFAULTAREA

Entering a variable name (for example, %Name) in the Cell Name field automatically enters the appropriate token in the "Select" field of the Content Query grid on the Display tab.

- **Tool Tip:** The text that appears as a tool tip when the cursor points at any data cell in the column. If this field is blank, the default Windows tool tip applies. For example, if the column width is reduced to where the text or value is not fully visible, the tool tip shows the complete text or value. If the column width is sufficient to show the full text or value, no tool tip is shown.
- **Action Left:** The action (Netview command, display command, or pop-up name) that is performed when the left mouse button is clicked on any data cell in this column. This action applies to the column at all times.

Note

The conditional actions that are defined for the column on the Multiple Items tab override the Action Left action.

- **Action Enter:** The action (Netview command, display command, etc.) that is performed when this cell is selected and the Enter key is pressed. The normal use for this function is in combination with cells that allow text entry.
- **Context Menu:** The name of the context menu (defined on the Context Menus tab) that contains the column-specific menu items of the grid tabular display.

The context menu appears when the user right-clicks on a cell in the grid tabular display. It includes the column-independent menu items of the context menu specified on the Display tab,

plus the column-specific menu items of the context menu specified on the Columns tab.

- **Visibility:** Name of the condition (defined on the Conditional Tests tab) that must be true for the value or image displayed in this cell to be visible. Only one conditional test can be selected from the drop-down list.

Note

The visibility condition defined here can be overridden by the definition on the Multiple Items tab. If there is a Multiple Items definition for the column, Visibility should be set to "None".

- **Logical Col Visibility:** The list of logical conditions that are used to determine the color of the cell value. All logical conditions defined on the Logical Conditions tab are listed here. Multiple logical conditions can be selected. The logical conditions can be toggled On/Off by clicking the condition name and clicking Apply.

Note

Logical Color Visibility has no effect if multiple items are defined for the column.

- **Total Type:** Defines the type of the totals cell of the column (for example, Text, Sum, or Average). Activated only for the following Placement Type values: Text, DateTime, and MultipleItems.
- **Total Text:** Specifies the numerical value or text string to be shown in the totals cell of the column. For the numeric totals type (for example, Sum), the text will be shown before the aggregated numeric value. Activated only when the Total Type field is specified.

Note

To show the totals row, specify the Totals Row value on the Display tab.

- **Regex:** The regular expression, which defines the allowable value format. Regular expressions may be used to validate field values that are manually entered by users.

2.4.3.2. Type Options

The Type group includes the following fields:

- **Placement Type:** The type of data displayed in this column.

Note

The tool automatically applies the Multiple Items placement type for columns that have conditions defined on the Multiple Items tab.

- **Text:** Character-based data that can be numerical values or text strings. When the text type is selected, an optional check box is displayed to allow the text field to be edited by the user.

The precision of numerical data is specified by the Precision Value field.

- **Text Edit Allowed:** Checking this box for a text placement type allows the user to edit the value displayed in the cell. This allows the user to input data on the tabular display. The cells where editing is allowed are indicated with a cyan background color.
- **Precision Value:** Sets the number of digits to the right of the decimal point that are displayed for numeric values. Note that the value displayed is rounded to achieve the required precision, not simply truncated.

3LIONS - Capacitor Input Display			
Name	ID	Nominal Hours Between Consecutive Switching	Manual Hours Between Consecutive Switching
1574	10218266	0.25	1.00
1590	10219078	0.25	0.00

Text placement (centered)

Text placement for a numeric value with a precision of 2

Text placement with edit allowed

- **Image:** A bitmap image (the default size is 24 by 24 pixels). The name of the image is defined in the Image field. The default folder for the location of all bitmap images used in the grid tabular displays is defined on the Display tab. The image below shows examples of bitmap placements. In many cases, a bitmap is used to indicate that clicking this cell navigates to another display.

- **Image:** The name of the image file to be displayed in the cell for a bitmap placement. To select an image file, click the Browse button to the right. The default directory for bitmap images used by the tabular displays is defined on the Display tab.

Station Feeder Input							
	Station	Feeder Input	Nominal Load Scaling Factor	Manual Load Scaling Factor	Manual LVM Maximum Power Factor Target Setting	Block All Regulating Transformers in the Station	Enable Voltage Reduction for all Transformers in the Station
 	3LIONS		1.00	--	--	☒	☐

Image placement (with Action)

Check Box placement

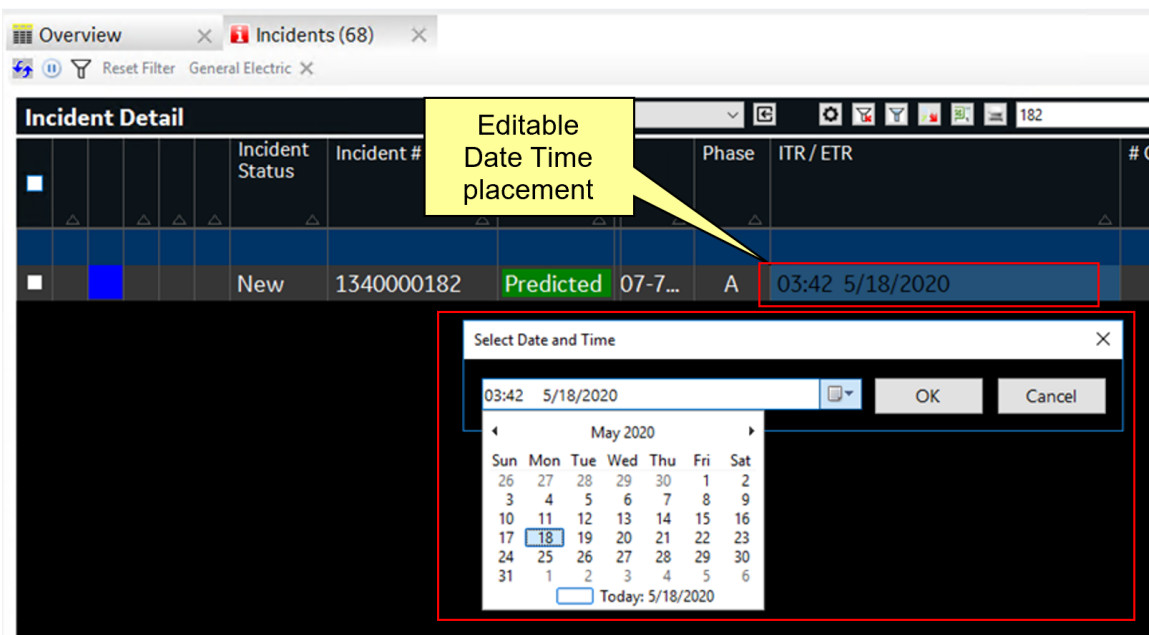
- **Check Box:** A single check box that the user can select or clear to enable/disable a function

or operation.

- **Code Translation:** Indicates that the display of the value is controlled by a coded translation item (image or short name can be shown). The details of the coded translation item are given in the Cell Name field.

Coded translation items are defined in the CodedTranslationConfig.xml file. This file can be viewed and edited via the Coded Translation Editor by selecting the Coded Translation Item from the pull-down list at the top of the page and viewing the parameters for each key value. For more information, refer to the *Coded Translation Editor User Guide*. For an example of a Grid Tabular display that uses coded translations, refer to the FindResultsGriddisplay.xml file.

- **Date Time:** Displays the Date and Time values returned by the data query. The image below shows an example of a Date Time placement.
 - **Precision Value:** Sets the date-time format.
 - **Allow Value Editing:** Specifies whether the date and time field can be manually edited by the user.



- **Check Box Single Select:** Multiple check boxes on the respective column, of which only one can be selected at a time (similar to radio buttons). Selecting a check box clears the previous selection.
- **Multiple Items:** Multiple items can be displayed in this cell, as defined on the Multiple Items tab (see [Multiple Items Tab](#)).
- **Check Box Multi-Select:** Multiple check boxes on the respective column, which can be used to select multiple rows (see [mulExample 4: Multi-Select Display](#)). The Select All check box is available in the column header.
- **Check Box Multi-Select no All:** Multiple check boxes on the respective column, which can

be used to select multiple rows (see [mulExample 4: Multi-Select Display](#)). The Select All check box is not available.

- **Check Box Single Row Select:** Multiple check boxes on the respective column, which can be used to select rows. Only one row can be selected at a time. Selecting a check box clears the previous row selection.
- **Tree View:** Hierarchical element tree, which can include organizational hierarchy levels (for example, service centers, substations, feeders, and devices), device types, and so on. The elements are listed in alphabetical order. For a configuration example, see [Example 6: Transformers Tree View Display](#).

The hierarchical element tree is configured using the Cell Name field. The Cell Name value format for the Tree View placement type is <Query_N>|<Query_X>.

For logical hierarchy levels, such as state, city, and service center, the %ORGLLEVEL_<N>_NAME and %ORGLLEVEL_<N>_ID queries are supported (for example, %ORGLLEVEL_1_NAME for the service center name). However, these queries are resolved to the associated values only when the showing the grid tabular display inside the Device Tree View popup.

For example, the Cell Name value

%ORGLLEVEL_2_ID|%ORGLLEVEL_1_NAME|%TOPOSUBNAME|%FEEDERNAME|%DNOMTABNAME|%ID defines that the hierarchical tree shows the DNOM Record Type, City, Service Center, Station, Feeder, and Device hierarchy levels.

- **Row Index:** Shows the row index or the sorting index.
 - When the Cell Name field is empty, the field shows the current row index (starting from 1)
 - When the Cell Name field is set to a sorting criteria in the format used by the Initial Sort field (for example, "CustOutCount:DESC|Priority:DESC"), the field shows the sorting index by the specified sorting criteria. For more details about specifying the Initial Sort Field, see [General](#).

This sorting index is not affected by the sorting criteria applied to the table statically or dynamically.

Note

The sorting index of the Row Index type is not recalculated when you apply table filtering. Use the Row Index Local type instead.

For a configuration example, see [Example 7: Display with a Row Index](#).

- **Row Index Local:** Shows the row index or the sorting index. This column type works similarly to the Row Index type, except that the sorting index is recalculated when you apply table filtering locally.

The following example shows row index, sorting index, and local sorting index columns, where the sorting index columns are sorted by the number of de-energized customers in

descending order (higher values first). In the example, the sorting index and local sorting index values differ because some entries were filtered out due to column-specific filtering.

Incident Detail				
Row Index	Sorting Index	Sorting Index Local	De-energized Customers	Station
			3	C
51				
1	20	3	3	COHO
2	8	2	13	COHO
3	7	1	35	COHO

2.4.3.3. Layout Options

Layout

Alignment:
Center

Default Width:
130
Minimum Width:
5

☐ Fix
☐ Hide
☒ Sortable
☐ Frozen

Update
Cancel

The Layout group includes the following fields:

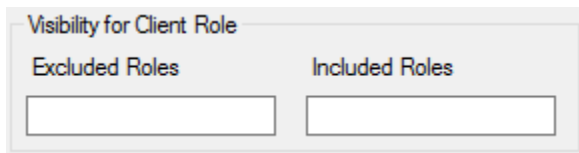
- **Alignment:** The justification of the information within the cell (i.e., left, right, or center).
- **Default Width:** The default width of the column (in pixels) when it is first displayed. If the Fixed Width parameter is not set, the user can adjust the width of the column from its default value.
- **Minimum Width:** The minimum width of the column (in pixels).

Note

The column width and order can be changed at run time. User preferences for modified displays are automatically stored in XML files and applied each time the user opens a display. The XML files with preferences are stored in the Grid_<Display_Name>.xml file located in the C:\Users\<User>\AppData\Roaming\GE\Browser directory.

- **Fix:** Select this check box to fix the width of the column. Column widths can no longer be changed on the display.
- **Hide:** Select this check box to hide this column from being displayed. This is useful when additional columns of data are required for conditional tests but are not part of the tabular display.
- **Sortable:** Select this check box to make this column sortable (i.e., the user can click in the header cell to sort the table by the column values). The columns that have the sort function enabled are indicated by an asterisk ("*") appended to the header text.
- **Frozen:** Select this check box to freeze the location of the column (i.e., the column cannot be dragged by the user to change its position within the display).

2.4.3.4. Client Role Visibility Options



The screenshot shows a dialog box titled "Visibility for Client Role". Inside the dialog, there are two labels: "Excluded Roles" and "Included Roles". Below each label is a text input field. The "Excluded Roles" field is on the left and the "Included Roles" field is on the right.

Use this group to define column visibility for different client roles (user types). This allows different columns to be shown for operators, dispatchers, analysts, and other users.

For more information about configuring user roles and running the ADMS client for a specific user role, refer to the *Viewer Configuration Editor User Guide*.

The Role Visibility group includes the following fields:

- **Included Roles:** A comma-separated list of roles for which the column will be shown. The column will not be shown for all the other roles.
- **Excluded Roles:** A comma-separated list of roles for which the column will not be shown. The column will be shown for all the other roles.

Note

You can specify either included or excluded roles for a column.

Caution

Role names in the Viewer Configuration Editor and Grid Tabular Builder must be consistent.

2.5. Conditional Tests Tab

Conditional tests allow the information in the tabular display to be modified based on the values of the data returned by the queries. The conditional tests are usually applied in visibility conditions.

Each conditional test can use up to four values (i.e., two tests between two pairs of values combined

by a logical operator). For example:

Result = (A compared with B) AND/OR (C compared with D).

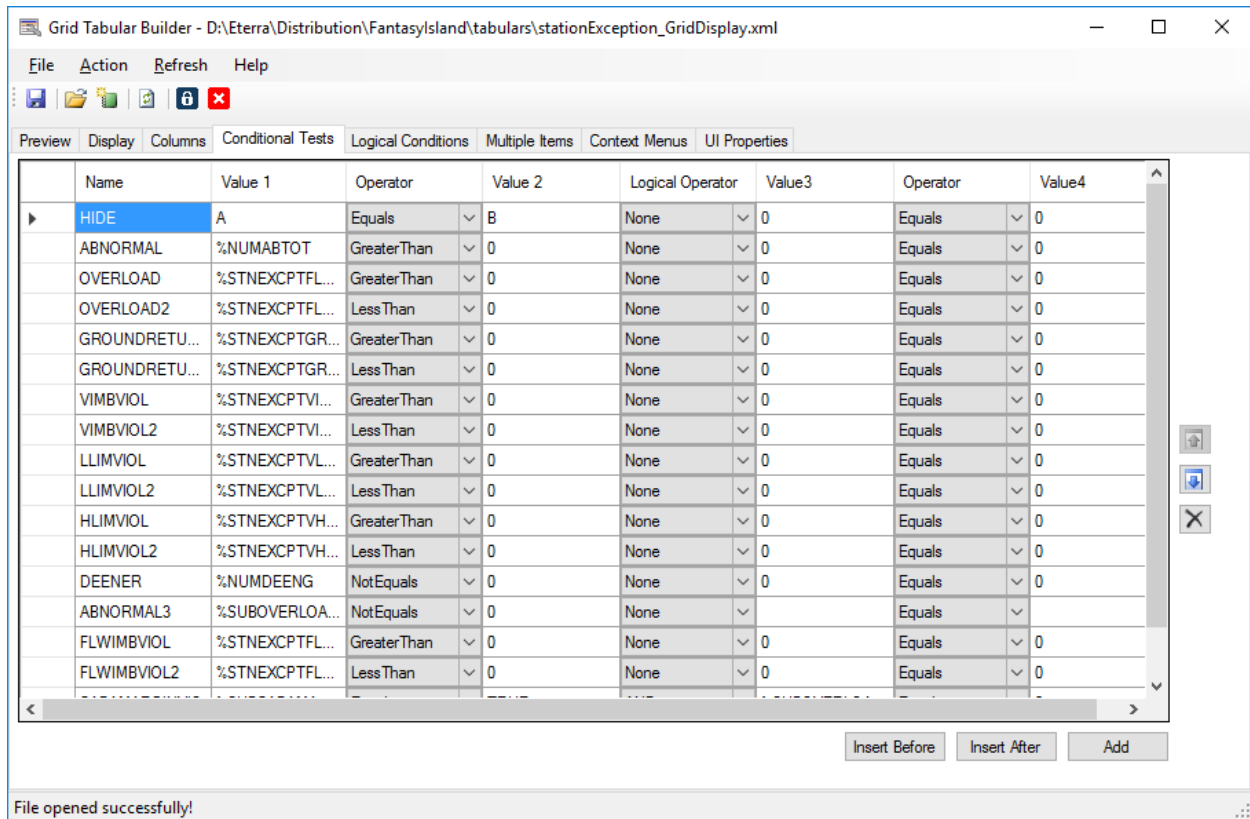


Figure 8. Conditional Tests Tab

The conditional tests can be edited directly on the Conditional Tests tab.

The Conditional Tests tab includes the following columns:

- **Name:** The name of the conditional test.
The conditional test names are used in visibility conditions on the Columns, Multiple Items, and Context Menus tabs. If the conditional test is True, the corresponding visibility condition is satisfied.
- **Value 1:** The first value used for the test. This can be a fixed value (text or numeric) or a value returned by a network model query.
- **Operator:** The operator to be applied for the test of Values 1 and 2:
 - Equals
 - Not Equals
 - Greater Than
 - Greater Than Equal (greater than or equal to)
 -

Less Than

- Less Than Equal (less than or equal to)
- **Value 2:** The second value, which is tested against Value 1.
- **Logical Operator:** The logical operator (AND/OR) that is used to combine the results of the two tests.
- **Value 3:** The third value, applied to the test of Values 3 and 4.
- **Operator:** The operator that is to be applied for the test of Values 3 and 4:
 - Equals
 - Not Equals
 - Greater Than
 - Greater Than Equal (greater than or equal to)
 - Less Than
 - Less Than Equal (less than or equal to)
- **Value 4:** The fourth value, applied to the test of Values 3 and 4.

The Conditional Tests tab includes the following buttons:

- **Insert Before:** Creates a new conditional test before the selected row.
- **Insert After:** Creates a new conditional test after the selected row.
- **Add:** Creates a new conditional test at the bottom of the list.
- **Move Up:** Moves the selected conditional test up one position.
- **Move Down:** Moves the selected conditional test down one position.
- **Delete:** Deletes the conditional test.

2.6. Logical Conditions Tab

The Logical Conditions tab is used to assign colors to the results of tests between two values.

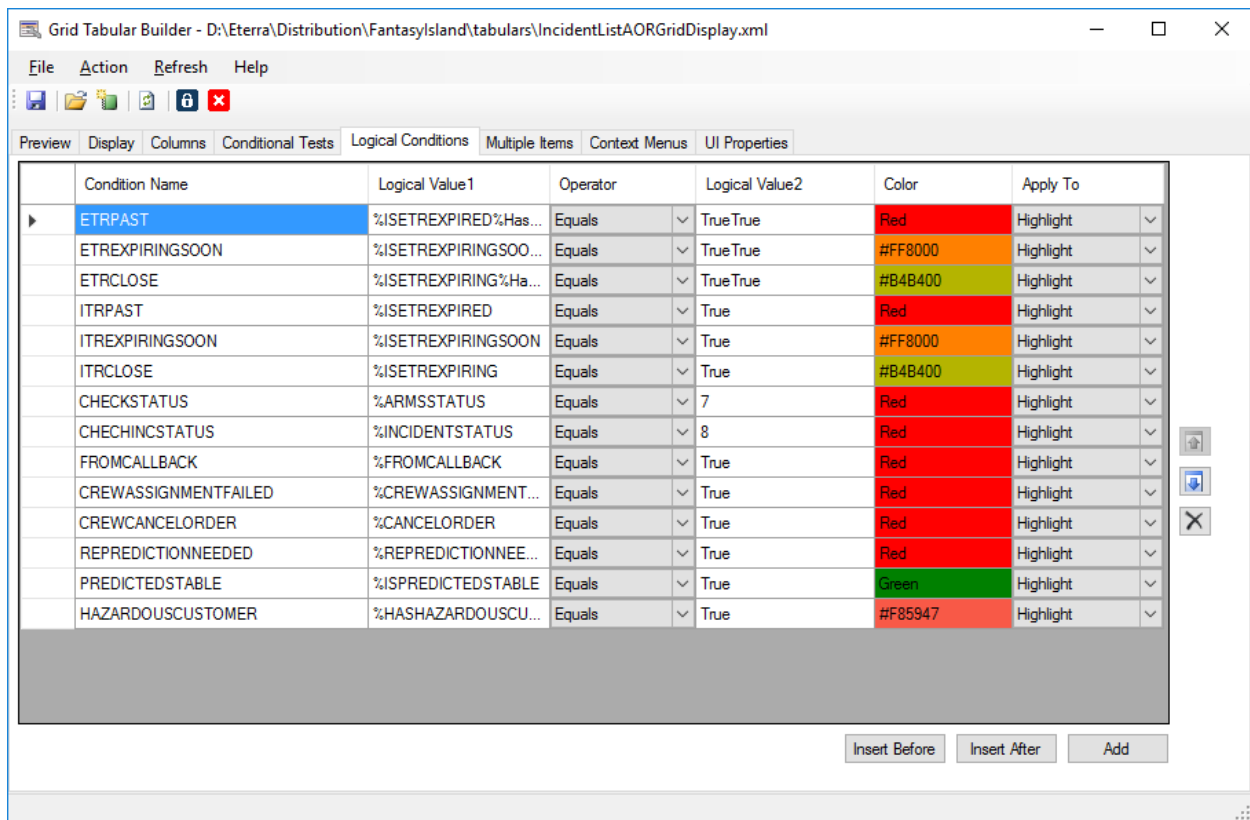


Figure 9. Logical Conditions Tab

The logical conditions are typically used to assign colors to values that are derived from queries or coded translations. The color is applied to the background of the cell (see following figure). In some cases, it may be more effective to define the colors for each state by using the Multiple Items tab.

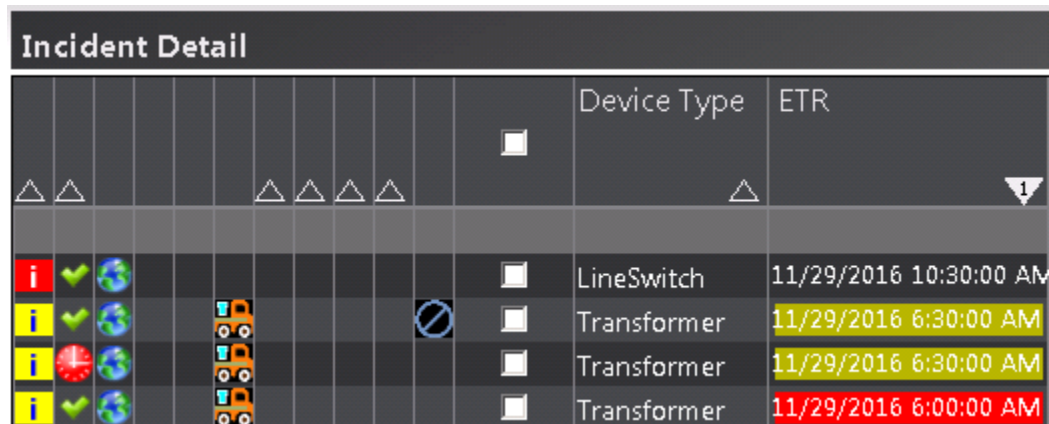


Figure 10. Logical Conditions Used to Control Text Highlight

The existing logical conditions can be edited directly on the page.

Each column on the Logical Conditions tab is described below:

- **Condition Name:** The name given to the logical condition — effectively, the result of the logical condition test.

- **Logical Value 1:** The first value used for the test. This can be a fixed value (text or numeric) or a value returned by a network model query.
- **Operator:** The operator to be used for the comparison test between the two logical values:
 - Equals
 - Not Equals
 - Greater Than
 - Greater Than Equal (greater than or equal to)
 - Less Than
 - Less Than Equal (less than or equal to)
- **Logical Value 2:** The second value used for the test. This can be a fixed value (text or numeric) or a value returned by a network model query.
- **Color:** The color that is to be assigned to the condition name if the result of the test is true.

Note

Logical condition colors do not take any effect if the multiple items capability (see [Multiple Items Tab](#)) is defined for the column.

- **Apply To:** Specifies the highlight type to be applied to the condition name.
 - **Background:** Changes the color of the entire cell.
 - **Text:** Changes the color of the cell text.
 - **Highlight:** Changes the color of the text highlight box inside the cell.
 - **None:** No color changes.

The Logical Conditions tab includes the following buttons:

- **Insert Before:** Creates a new logical condition before the selected row.
- **Insert After:** Creates a new logical condition after the selected row.
- **Add:** Creates a new logical condition at the bottom of the list.
- **Move Up:** Moves the selected logical condition up one position.
- **Move Down:** Moves the selected logical condition down one position.
- **Delete:** Deletes the currently selected logical condition.

2.7. Multiple Items Tab

The Multiple Items tab allows more than one information item to be displayed within a single cell. The determination of which item is displayed in the cell is based on conditional tests.

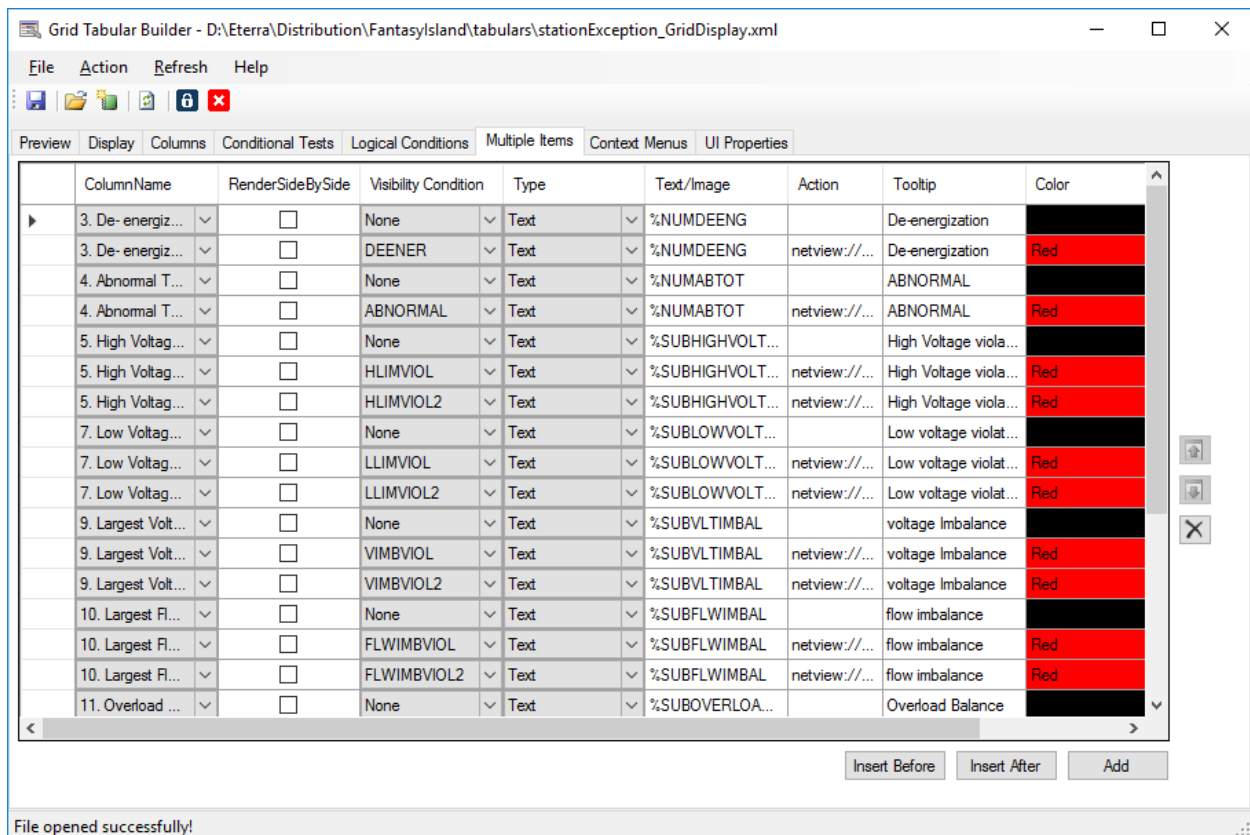


Figure 11. Multiple Items Tab

The multiple items capability allows for greater flexibility in grid tabular displays. It can be used to change the color of text, or to display a combination of text and bitmap images to identify various conditions.

Figure below shows an example of using multiple items, where the value is shown with a red highlight if it is non-zero. This is done by defining two separate items (the value with no highlight and the value with red highlight) and then using a conditional test to determine which item is displayed.

Station Exception				
Station	Area	Locate	De- energized Customers	Abnormal Topology Conditions
<u>ROYAL TROON</u>	WST		5186	1
<u>3LIONS</u>	WST		425	1

Figure 12. Example of a Multiple Items Display

The Multiple Items tab includes the following columns:

- **Column Name:** The column where the information item is to be displayed. A separate entry is required for each information item that is displayed in the column. Therefore, there is more

than one row on the Multiple Items tab for each column of the display.

- **Render Side by Side:** Checking this box displays the information items side by side within the cell rather than over the top of each other. This can allow more than one value to be displayed in a single cell.
- **Visibility Condition:** The conditional test that causes the information item to be displayed. If the result of the conditional test is true, this information item is displayed in the cell.

For each Multiple Items column, there must be a default condition of "None", which defines the information display when none of the other visibility conditions are true.

- **Type:** The information item display type: Text or Image.
- **Text/Image:** The data value or image file name that is displayed for this information item. In the case of images, the default directory for the image files is defined on the Display tab.
- **Action:** The action (Netview command) that is performed if this information item is selected (with either mouse button). In this way, a different action can be defined for each information item. Note that this action overrides any actions that can be defined on the Columns tab.
- **Tool Tip:** The tool tip text that is displayed when the cursor points at this information item. In this way, a different tool tip can be displayed for each information item. Note that this tool tip overrides any tool tip defined on the Columns tab.
- **Color:** The color used to highlight this information item. Select the desired cell to display the color selection palette. Both basic colors and custom colors can be selected. Custom colors are shown in the cell as their numeric value rather than the actual color; however, the correct color appears in the grid tabular display.
- **Apply To:** Specifies the highlight type to be applied to this item:
 - **Background:** Changes the color of the entire cell.
 - **Text:** Changes the text color within a cell.
 - **Highlight:** Changes the color of the text highlight box inside the cell.
 - **None:** No color changes.

The Multiple Items tab includes the following buttons:

- **Insert Before:** Creates a new multiple condition item before the selected row.
- **Insert After:** Creates a new multiple condition item after the selected row.
- **Add:** Creates a new multiple condition item at the bottom of the list.
- **Move Up:** Moves the selected multiple condition item up one position.
- **Move Down:** Moves the selected multiple condition item down one position.
- **Delete:** Deletes the currently selected multiple condition item.

2.8. Context Menus Tab

The Context Menus tab allows you to define context menus for the grid tabular display. You can define both the column-independent (common) context menu and the column-specific context menus.

The context menu is displayed when you right-click on a cell in the tabular display. It includes the items of the common menu plus the items of the column-specific menu. The common menu is specified on the Display tab and the column-specific menu is specified on the Columns tab.

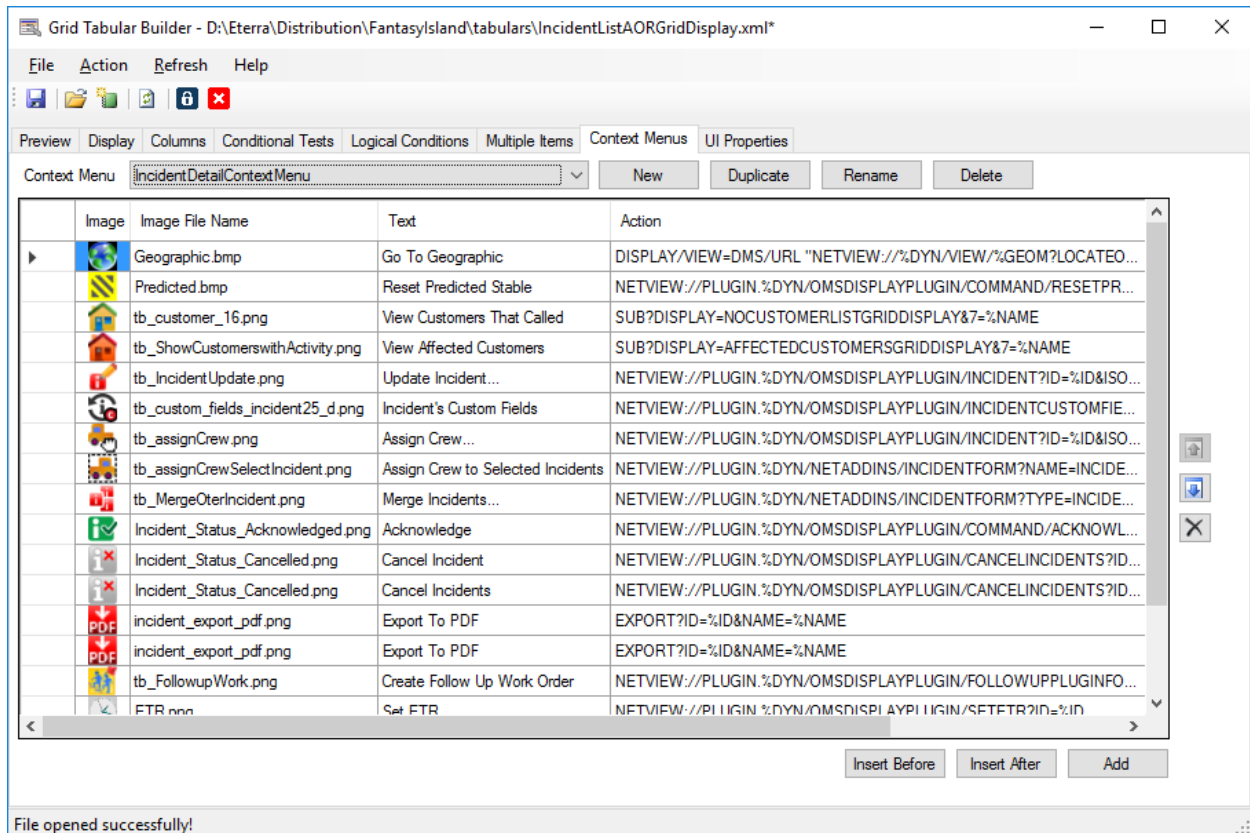


Figure 13. Context Menus Tab

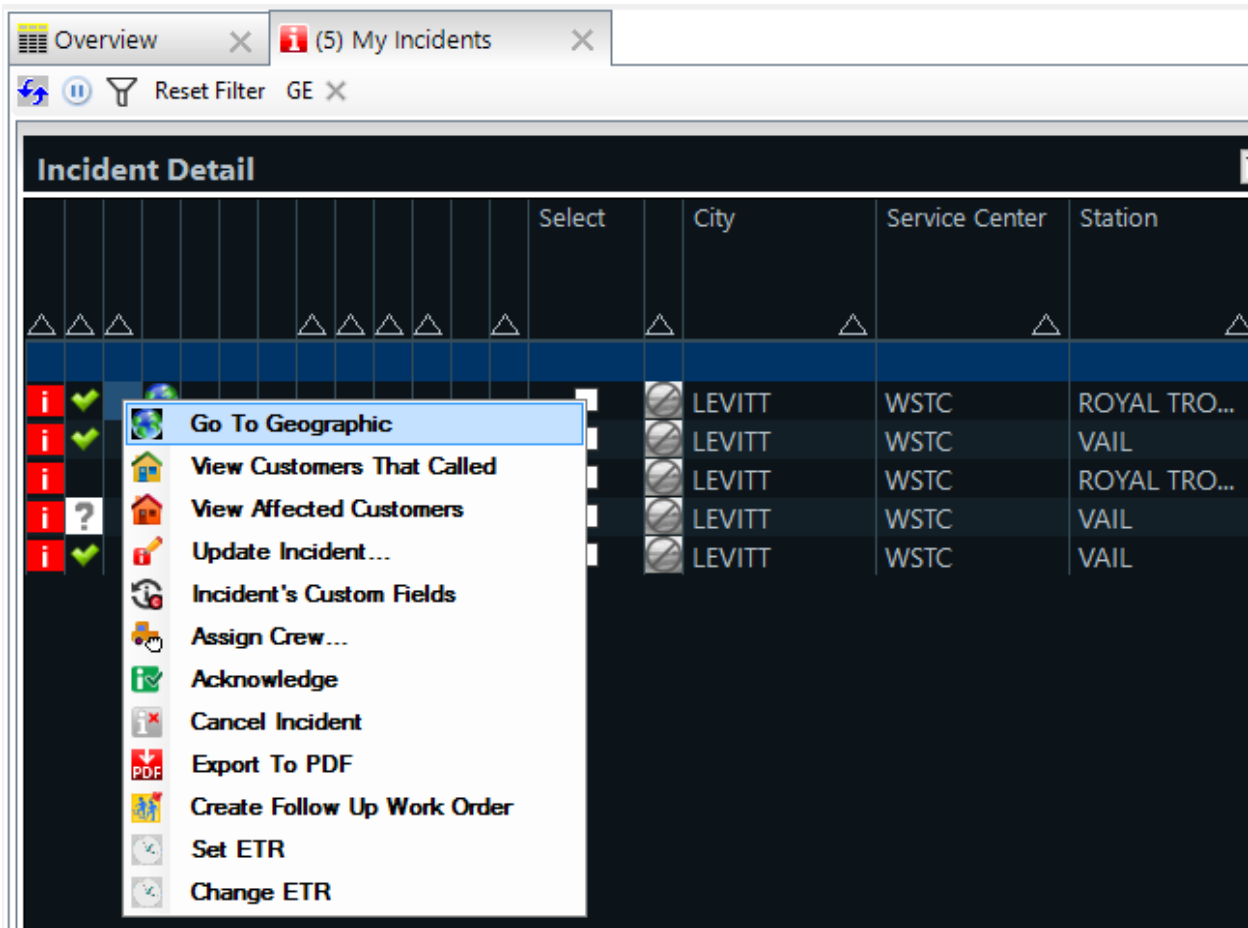


Figure 14. Example of a Context Menu

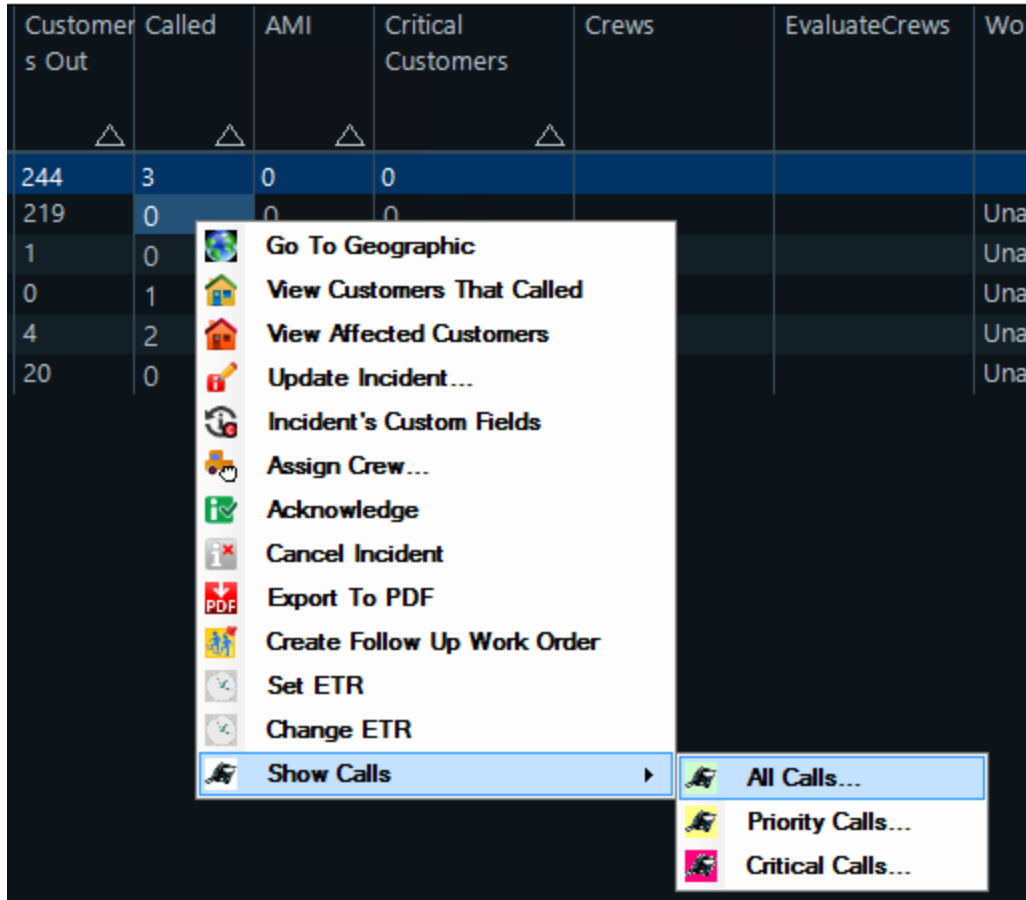


Figure 15. Example of a Context Submenu

2.8.1. Context Menu Tab Items

Each row in this grid tabular control represents one context menu item.

The Context Menu tab includes the following controls:

- **Context Menu:** Displays a drop-down list of context menus defined for the current display. By default, the first context menu is displayed.

To view or edit the context menu items for another context menu, select the desired context menu name.

A context menu name references a set of context menu items. The name does not appear in the context menu, but it is referenced when specifying a column-independent (common) context menu on the Display tab, or a column-specific context menu on the Columns tab. It is also referenced in the Action column of the Context Menus tab when defining submenus, which are menu items that are themselves menus.

- **New:** Creates a new blank context menu.

To create a new context menu without menu items, click the New button, and then enter the

name in the Context Menu Name pop-up. This causes the Context Menu Name field to be set to "<new context menu name>", and the context menu items are cleared.

- **Duplicate:** Creates a context menu similar to an existing one.

To create a new context menu with the same menu items as the current context menu, click the Duplicate button, and then enter the name in the Context Menu Name pop-up. This causes the Context Menu Name field to be set to "<new context menu name>", and the context menu items are initialized with the same menu items that were defined for the previous context menu. This simplifies creation of similar context menus, since it is easier to delete or change menu items than to create them.

- **Rename:** Renames the context menu.

To rename a context menu, click the Rename button, and then enter the new name in the Context Menu Name pop-up.

- **Delete:** Deletes the context menu. Note that this deletes the entire context menu, not just one context menu item.

The Context Menu tab includes the following columns:

- **Image:** Context menu item image.

Clicking the image opens a window to select another image.

- **Image File Name:** Context menu item image file name.

To select another image, enter the name of the new image in this column.

- **Text:** Context menu item text.

- **Action:** Action to be performed when this context menu item is selected. The action is specified as a URL (for a description of action URLs, see [Action URLs](#)).

To insert a menu item separator (a horizontal bar that visually separates menu items into groups), enter either the word "SEPARATOR" or a dash ("-") in either the Text or Action column.

To specify that the context menu item is a submenu, enter "Submenu:" followed by the name of the context menu that contains the submenu items. For example, if the action for the "Show Calls" menu item is "Submenu:ShowCallsSubmenu", when the user points to the "Show Calls" menu item, a cascading submenu, ShowCallsSubmenu, with additional menu items is displayed. The submenu need not be defined before it is referenced in this way, but it must be defined before the context menu can work as desired on a grid tabular display.

- **Use In Multi-Select:** Option to display a context menu item in multi-select mode. For example, if you select more than one row in a grid and right-click on the selection, the context menu will display only the items with the "Use In Multi-Select option" flag. All other context menu items are inaccessible.
- **Visibility Condition Name:** Name of the condition (defined on the Conditional Tests tab) that must be true in order for this context menu item to be visible. This enables you to control whether a context menu item appears based on some condition that is evaluated for the

selected row of the grid tabular display.

For example, the View Assigned Crew item is visible only if the condition "IsCrewAssigned" is true. Therefore, if a crew is not assigned to an incident, the View Assigned Crew item does not appear in the context menu for that incident.

The Context Menu tab includes the following buttons:

- **Insert Above:** Adds a context menu item above the selected row.
- **Insert After:** Adds a context menu item below the selected row.
- **Add:** Adds a context menu item at the end of the list.
- **Move Up:** Moves a selected menu item up one row.
- **Move Down:** Moves a selected menu item down one row.
- **Delete:** The bottom button to the right of the Context Menu Items grid.

2.9. UI Properties Tab

The appearance of all grid tabular displays is specified in the client configuration file, NetConf.xml, which can be configured by the Net Dynamics Configuration Editor. For more details, refer to the *Net Dynamics Configuration Editor User Guide*.

On the UI Properties tab, you can apply custom UI settings for grid tabular displays.

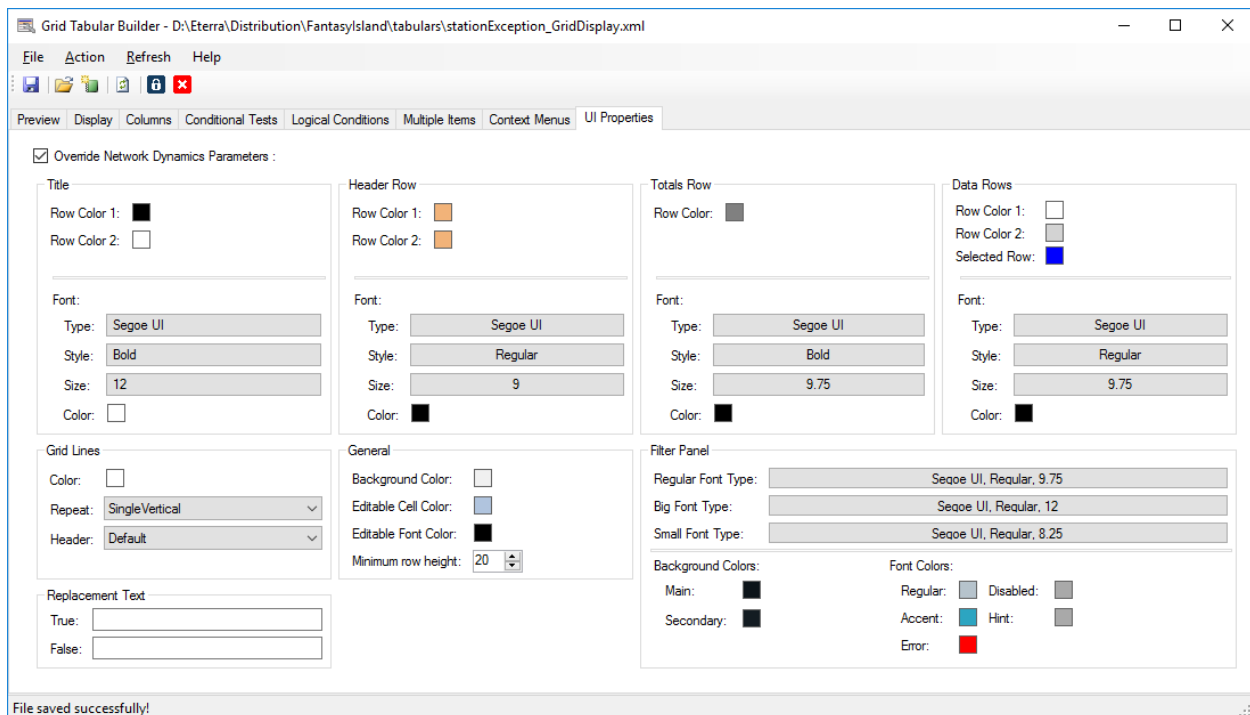


Figure 16. UI Properties Tab

The UI Properties tab includes the following controls:

- **Override Network Dynamics Parameters:** Enables or disables the custom UI parameters for the grid tabular display.

Note

By default, this option is not selected and the default grid tabular UI parameters are used as specified in the client configuration file.

- **Title:** Color settings for the title row.
 - **Row Color 1:** Sets the start color for the display title's background gradient. The title gradient is painted left-to-right using the Row Color 1 and Row Color 2 colors.
 - **Row Color 2:** Sets the end color for the display title's background gradient. colors.
 - **Font:** The font settings (type, style, size, and color) for the title.
- **Header Row:** Color settings for the header row.
 - **Row Color 1:** The background color of the header column for odd-numbered columns.
 - **Row Color 2:** The background color of the header column for even-numbered columns.
 - **Font:** The font settings (type, style, size, and color) for the header row.
- **Totals Row:** Color settings for the totals row.
 - **Row Color:** The background color of the totals row.
 - **Font:** The font settings (type, style, size, and color) for the totals row.
- **Data Rows:** Color settings for data rows.
 - **Row Color 1:** The background color of the data row for odd-numbered rows.
 - **Row Color 2:** The background color of the data row for even-numbered rows.
 - **Highlighted Row:** The background color of the selected data cell row.
 - **Font:** The font settings (type, style, size, and color) for the data row.
- **Grid Lines:** Style and color of grid lines.
 - **Color:** Defines the color for grid lines.
 - **Repeat:** Defines the grid tabular border style. The following options are available:
 - Single: A single line border.
 - Raised: A three-dimensional raised border.
 - Sunken: A three-dimensional sunken border.
 - None: No borders.
 - SingleVertical: A vertical single-line border.
 - RaisedVertical: A vertical three-dimensional raised border.
 - SunkenVertical: A vertical three-dimensional sunken border.
 - SingleHorizontal: A horizontal single-line border.
 -

RaisedHorizontal: A horizontal three-dimensional raised border.

- **SunkenHorizontal:** A horizontal three-dimensional sunken border.
- **Header:** Defines the header line style. The following options are available:
 - **Default:** Default line style.
 - **None:** No border.
 - **Single:** A single line border.
 - **SingleVertical:** A vertical single-line border.
 - **SingleHorizontal:** A horizontal single-line border.
- **Replacement Text:** Defines the replacement values for the True and False flags (for example, Yes and No).
- **General:** General style settings.
 - **Background Color:** Defines the background color used behind the grid tabular display.
 - **Editable Cell Color:** Defines the background color of an editable cell.
 - **Editable Font Color:** Defines the text color of an editable cell.
 - **Minimum Row Height:** Defines the minimum row height (in pixels).
- **Filter Panel:** Grid tabular enhanced filtering panel settings. To enable enhanced filtering, set the Grid Tabular Enable Dynamic Filters option to True in the Net Dynamics Configuration Editor.
 - **Regular Font Type:** Defines the font type, style, and size for regular text.
 - **Big Font Type:** Defines the font type, style, and size for big text.
 - **Small Font Type:** Defines the font type, style, and size for small text.
 - **Background Main Color:** Defines the main background color for the filtering panel. To show the color selection palette, click the color square.
 - **Background Secondary Color:** Defines the secondary background color for the filtering panel.
 - **Regular Font Color:** Defines the regular font color for header rows on the filtering panel.
 - **Accent Font Color:** Defines the accent font color for the filtering panel.
 - **Error Font Color:** Defines the font color for erroneous values on the filtering panel.
 - **Disabled Font Color:** Defines the font color for disabled controls on the filtering panel.
 - **Hint Font Color:** Defines the font color for hints on the filtering panel.

Note

The grid tabular display user interface was significantly changed in **e-terr**distribution 3.5 (see figure below). To restore the legacy grid tabular appearance, select the Override Network Dynamics Parameters option for all displays, or modify the display settings in the client configuration file.

Station Exception								
Station	Area	Locate	De- energized Customers	Abnormal Topology Conditions	High Voltage Violation	Highest Load or Bus Voltage (%)	Low Voltage Violation	Lowest Load or Bus Voltage (%)
FISHER			31	0	No	100.00	Yes	34.37
CORTLAND			23	0	No	100.00	Yes	50.35
MILLER			9	0	No	100.00	Yes	53.10
STARR			7	0	No	100.00	Yes	23.33
PEBBLE BEACH	3LIONSSC		2	1	Yes	106.34	Yes	87.99
LABRADOR			2	0	No	100.23	Yes	57.53
RAVEN	3LIONSSC		1	1	No	103.56	No	94.55

Figure 17. Current grid tabular display appearance

Station Exception								
Station	Area	Locate	De- energized Customers	Abnormal Topology Conditions	High Voltage Violation	Highest Load or Bus Voltage (%)	Low Voltage Violation	Lowest Load or Bus Voltage (%)
FISHER			31			100.00	Yes	34.37
CORTLAND			23	0	No	100.00	Yes	50.35
MILLER			9			100.00	Yes	53.10
STARR			7	0	No	100.00	Yes	23.33
PEBBLE BEACH	3LIONSSC		2	1	Yes	106.34	Yes	87.99
LABRADOR			2	0	No	100.23	Yes	57.53
RAVEN	3LIONSSC		1	1		103.56		94.55
BETHPAGE	NEAC		1	0	No	103.61	Yes	94.79
SPYGLASS	STHC		1			105.06		96.84
NEPTUNE	NEAB		0	0	Yes	126.91	No	97.18
MUIRFIELD	NEAC				Yes	108.34	Yes	91.58
VAIL	CTLA		0	0	No	103.26	Yes	92.74
PINEHURST	STHB					104.87	Yes	94.56
JUPITER	CTLB		0	0	No	102.86	Yes	94.90
HEAVENLY	CTLA					102.89		95.95
PINE VALLEY	STHA		0	0	No	103.48	No	97.41

Figure 18. Grid tabular display appearance in versions 3.4 and earlier

3. Working with the Grid Tabular Builder

3.1. Starting the Grid Tabular Builder

To start the Grid Tabular Builder:

- From the Windows Start menu, point to Programs > Eterra > Distribution > Tools > Configuration Tools > Grid Tabular Builder.

The Grid Tabular Builder opens in a blank state (i.e., with no grid tabular display loaded). It is possible to immediately start building a new display. More frequently, however, an existing display is opened to modify it or use it as the starting point for a new display.

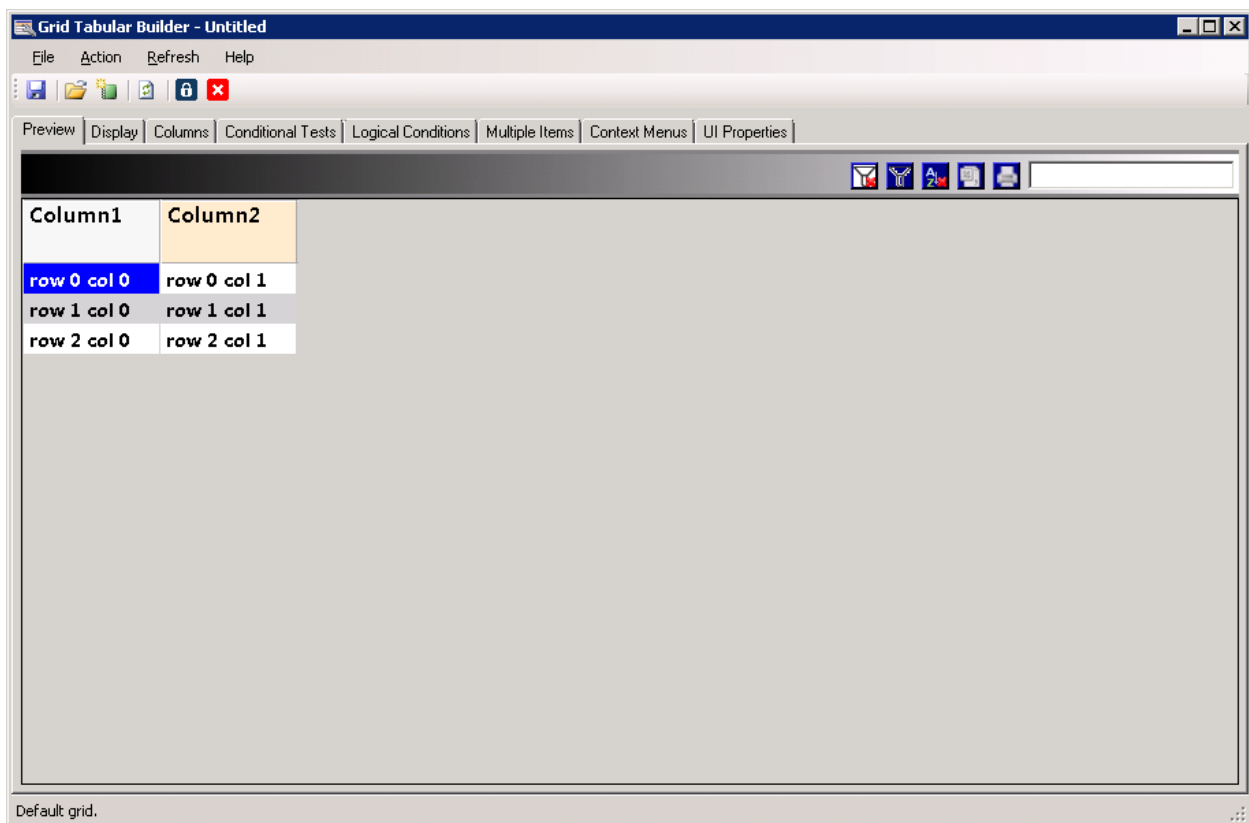


Figure 19. Grid Tabular Builder - Blank Display

3.2. File Locations and Conventions

The grid tabular display files (xxx_griddisplay.xml) are located in the ADMS "tabular" folder. The bitmap images used in the displays are located in the ADMS "bitmaps" folder. The default locations of these folders are:

- %IDMS_ROOT%/tabulars

- %IDMS_ROOT%/bitmaps

where %IDMS_ROOT% is the environment variable that points to the main path for ADMS folders.

The location of the bitmaps folder is defined in the Grid Tabular Builder startup parameter "-bpath <bitmap path>", where <bitmap path> is the full path to the folder.

It is important that these locations are used to ensure that all displays are recognized by the client. The locations of these folders are configurable in the Net Dynamics Configuration Editor.

3.3. Integrating Grid Tabular Displays

Critical parts in the development of tabular displays are the location and identification of the Distribution Network Object Model (DNOM) data values that are to be displayed.

The following ADMS product documents are the key references that describe the details of DNOM data and display commands and parameters:

- *ADMS Modeling Reference Guide*
- *ADMS Modeling Reference Tables*

Other useful sources of information about DNOM data and display commands are the existing displays provided in the standard ADMS product release.

3.3.1. DNOM Mapped Fields, Queries, and Query Helpers

For a detailed list of mapped fields and queries arranged by class, refer to the *ADMS Modeling Reference Guide*.

3.3.2. Existing Displays

Samples of key displays are described in [Standard Grid Tabular Displays](#).

For descriptions of all grid tabular displays, refer to the *ADMS Display Reference Manual*.

3.4. Creating a Tabular Display

The following examples show screenshots and explanations of how a range of displays are created and how the elements are constructed, along with explanations of how the displays are linked to the rest of the system. The example displays are simplified versions of displays that are contained in a standard ADMS release.

The example displays are:

- [Example 1: Station/Area List Display](#)

- [Example 2: Station/Abnormal List Display](#)
- [Example 3: Multi-Pane Display](#)
- [Example 4: Multi-Select Display](#)
- [Example 5: Auto-Refreshable Display](#)

For descriptions of URLs that specify the action to be taken when the user clicks an item associated with that action, refer to [Action URLs](#).

3.4.1. Example 1: Station/Area List Display

The Station/Area List display shows all the stations that are defined in the network model, bitmap links to the geographic network view centered on each station, and the Areas of Responsibility for each station.

Station/Area List		
		Areas
		Area: 3LIONS
		Area: BACHELOR
		Area: BAKER
		Area: BETHPAGE
		Area: CANTERBURY
		Area: COHO
		Area: CURLEW
		Area: DURBAN
		Area: EQSI
		Area: HEAVENLY
		Area: JUPITER
		Area: MARS
		Area: MERCURY
		Area: MOBILE1
		Area: MOBILE2

Figure 20. Station/Area List Display

The *ADMS Modeling Reference Guide* shows the list of attributes for the .SUB class.

The required attributes for the Station/Area List display are:

- **NAME:** Station display name
- **AREANAME:** Associated operation area/division
- **DYN:** The current dynamic instance

To create a Station/Area List Display:

1. Open the Grid Tabular Builder.
2. Save the blank display as StationAreaList_Demo_GridDisplay.xml in the default display folder (normally %IDMS_ROOT%\tabulars).
3. To prepare the file for further editing, click the Read Only button on the toolbar.
4. Select the Columns tab.
5. Create the first (left) column, named "Station", by clicking the Add button, and then enter the data as shown in the figure below:

Edit existing column

Field

Name:

Header Cell Name:

Cell Name:

ToolTip:

Action Left:

Action Enter:

Context Menu:

Visibility:

Logical Col Visibility:

Total Type:

Type

Placement Type:

☐ TextEditAllowed

Precision Value:

Layout

Alignment:

Default Width:

☐ Fix ☐ Hide ☒ Sortable ☐ Frozen

6. To add the information to the Columns tab, click the Update button.
7. Create the next column, named "Geo View", by clicking the Add button.
8. Open the Edit form and enter the information as shown in the figure below:

The important item for defining this column is the command in the Action Left text box, which is linked to the column.

The Netview command for displaying a geographic network view is:

```
NETVIEW://dynamics/VIEW/geometry/station_name?argument
```

In this case, the default arguments are used and the command becomes:

```
NETVIEW://%DYN/VIEW/DETAIL/%NAME
```

9. Select the Image option from the Placement Type drop-down list, and then select the Geographic.bmp image from the bitmap folder that is specified on the Display tab.
10. Create the third column, named "Areas", by entering the information shown in the figure below:

Edit existing column

Field

Name:

Header Cell Name:

Cell Name:

ToolTip:

Action Left:

Action Enter:

Context Menu:

Visibility:

Logical Col Visibility:

Total Type:

Type

Placement Type:

☐ TextEditAllowed

Precision Value:

Layout

Alignment:

Default Width:

☐ Fix ☐ Hide ☐ Sortable ☐ Frozen

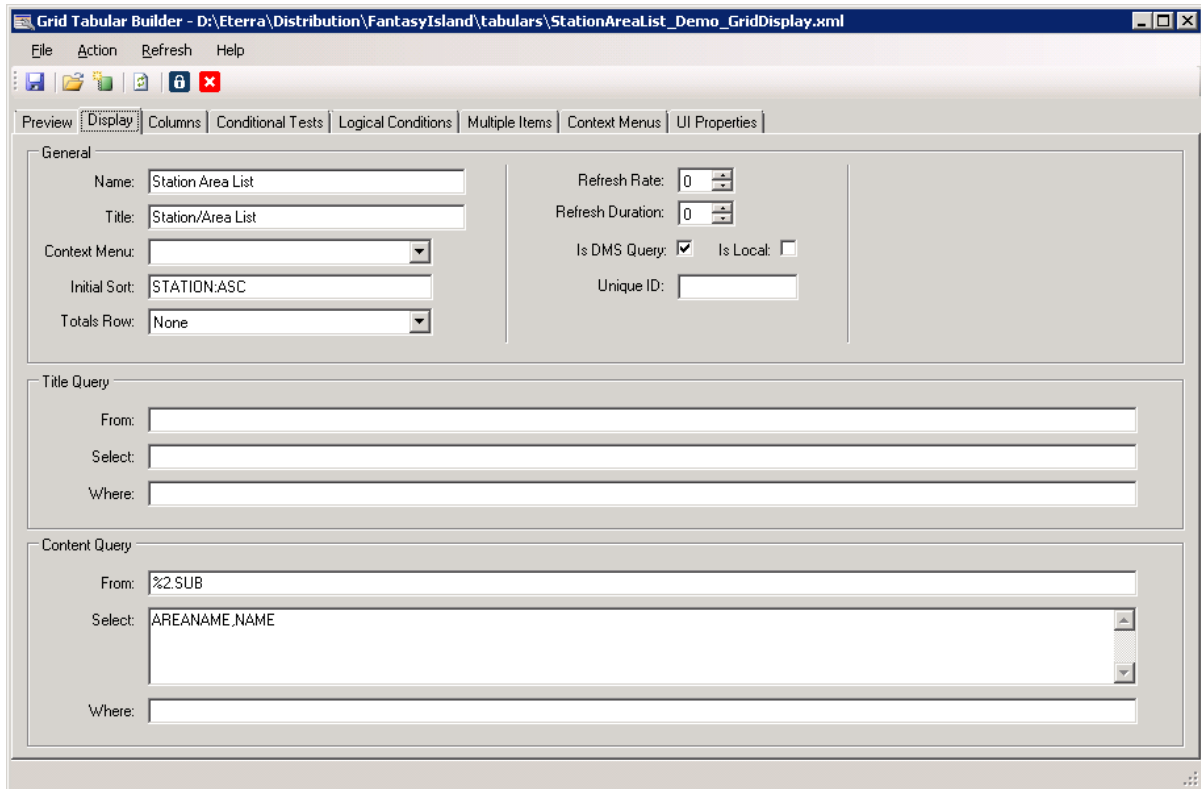
The display in each cell includes some fixed text as well as variable text. Therefore, the Cell Name field is:

Area: %AREANAME

11. On the Grid Tabular Builder, select the Display tab.

The Content Query Select field is already populated with the tokens from the column cell names (NAME,AREANAME).

12. Enter the remaining details on the page, as shown in the figure below:



The important item on this page is the query definition for the data that is repeated in the data cells.

From the *ADMS Modeling Reference Guide*, you know that the class for a substation is .SUB. The display must reflect the correct dynamic state; therefore, the FROM clause is:

```
%2.SUB
```

The %2 parameter specifies the dynamic state because it is the second argument ("RT") in the Netview command to call up the display.

```
NETVIEW://RT/gridtabular/StationAreaList_Demo_GridDisplay
```

13. Define the initial sorting method of the display to list the station names in ascending order by adding "%NAME:ASC" to the Initial Sort field.
14. Save the file.
15. Test the display by entering the Netview command in the text entry field on an ADMS viewport:

```
NETVIEW://RT/gridtabular/stationarealist_demo_griddisplay
```

3.4.2. Example 2: Station/Abnormal List Display

Building on the Station/Area List display, this example adds an additional column to display the

number of abnormal switches at each station.

If the abnormal value is zero, an image is displayed. If the abnormal value is greater than zero, the value is displayed in red, and a link to a detailed display of abnormal devices for the station is enabled.

1. In the Grid Tabular Builder, open the StationArea_Demo_GridDisplay.xml file and re-save it as StationAbnormal_Demo_GridDisplay.xml.

The number of devices that belong to a substation that are currently in an abnormal state is an attribute of the Substation class (NUMABTOT). For more information, refer to the *ADMS Modeling Reference Guide*.

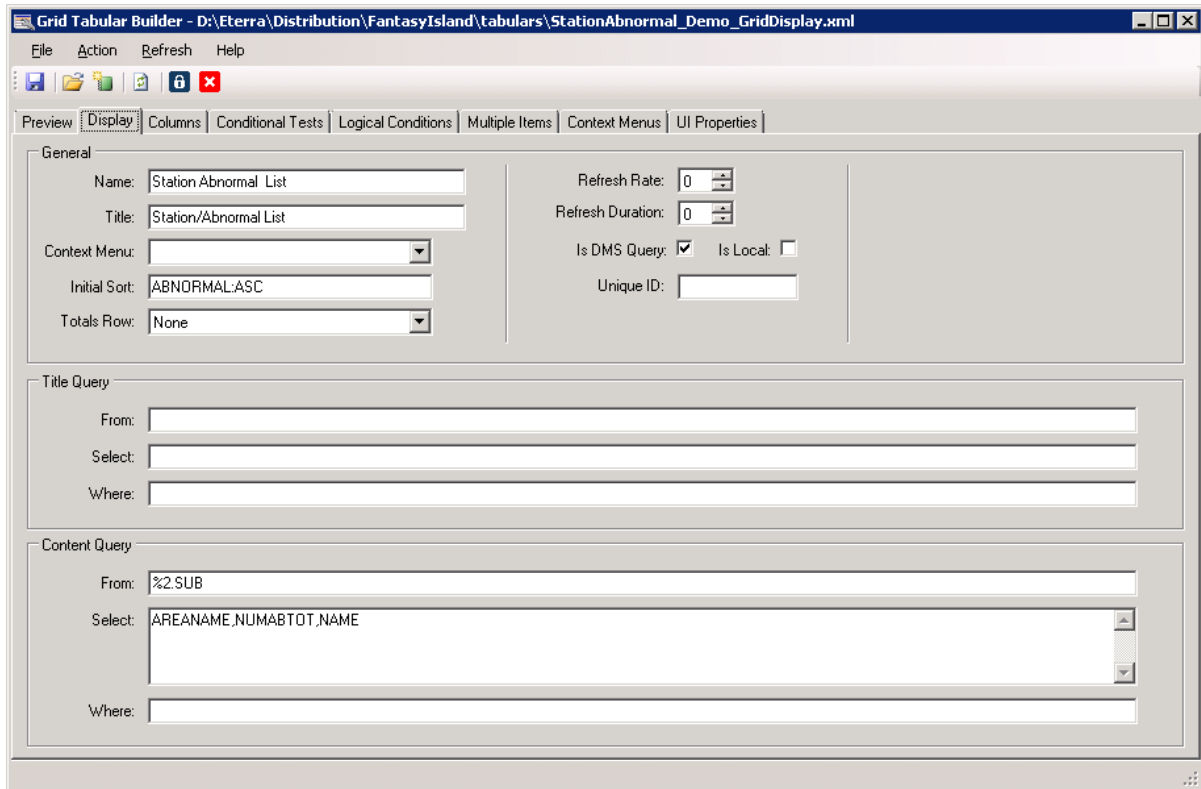
2. Open the Columns tab and add a new column, named "Abnormal", as shown in the figure below:

The screenshot shows the 'Edit existing column' dialog box. The 'Field' section on the left includes the following fields: Name (Abnormal), Header Cell Name (ABNORMAL), Cell Name (%NUMABTOT), ToolTip, Action Left, Action Enter, Context Menu, Visibility (None), Logical Col Visibility, and Total Type (None). The 'Type' section on the right includes a Placement Type dropdown set to 'Multiple Items' and a Precision Value field. The 'Layout' section at the bottom right includes an Alignment dropdown set to 'Center', a Default Width field set to '100', and checkboxes for Fix, Hide, Sortable (checked), and Frozen. At the bottom right are 'Update' and 'Cancel' buttons.

The placement type for this column is "Multiple Items", and the cell name is the number of abnormal devices: %NUMABTOT.

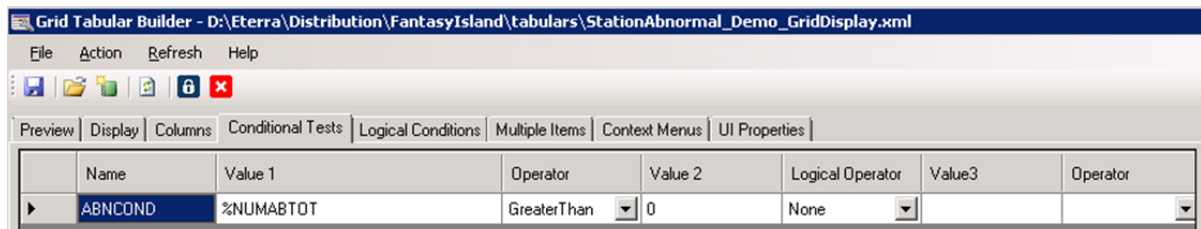
3. Open the Display tab, and ensure that NAME ,AREANAME ,NUMABTOT has been added to the Content Query SELECT clause.
4. Change the initial sort to show the stations with abnormal devices at the top by entering ABNORMAL : ASC.
5. Change the name and title of the display.

The updated Display tab is shown in the figure below:



6. Open the Conditional Tests tab and create a conditional test with the following properties:

- **Name:** ABNCOND
- **Value 1:** %NUMABTOT
- **Operator:** Greater Than
- **Value 2:** Zero

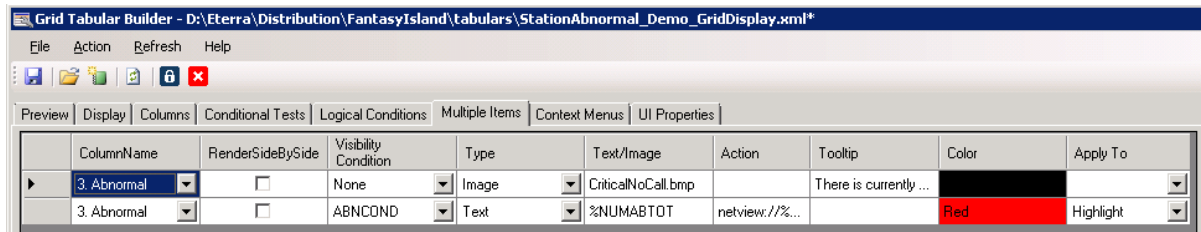


The result of this conditional test is used to trigger the appropriate display item for the Abnormal column. The display items are defined on the Multiple Items tab.

7. On the Multiple Items tab, create two items for the Abnormal column with the following properties:

- **None:** The item to be displayed when there is no defined condition. This item is defined as a bitmap image, a star symbol (file name CriticalNoCall.bmp).
-

ABNCOND: The item to be displayed when this conditional test is true. In this case, the value of %NUMABTOT is displayed in red, and the following action is linked to the cell: netview://%DYN/griddtabular/abnormal_griddisplay/%NAME



This Netview command displays a tabular summary of all the abnormal devices for the nominated station.

8. Save the display file.
9. Within the ADMS client, open the file using the command:

```
NETVIEW://RT/griddtabular/StationAbnormal_demo_GridDisplay
```

10. Verify that the display looks similar to the figure below and works as expected.

Station/Abnormal List				
Station	Geo View	Areas	Abnormal	
3LIONS		Area: 3LIONS	1	
CANTERBURY		Area: CTL	2	
RAVEN		Area: WST	2	
ROYAL TROON		Area: WST	2	
WINGED FOOT		Area: WST	2	
COHO		Area: WST	3	
BACHELOR		Area: NEA		
BAKER		Area: CTL		
BETHPAGE		Area: NEA		
CURLEW		Area: STH		
DURBAN		Area: STH		
EQSI		Area: SOUTH		
HEAVENLY		Area: CTL		
JUPITER		Area: CTL		
MARS		Area: STH		
MERCURY		Area: CTL		
MOBILE1		Area:		
MOBILE2		Area:		
MUIRFIELD		Area: NEA		

3.4.3. Example 3: Multi-Pane Display

Multi-pane displays, such as the Distribution Power Flow display that is accessed from the Station Output Summary display (see figure below), can be a very effective means of showing a diverse range of information on a single screen. A maximum of three frames can be defined for a single display screen.

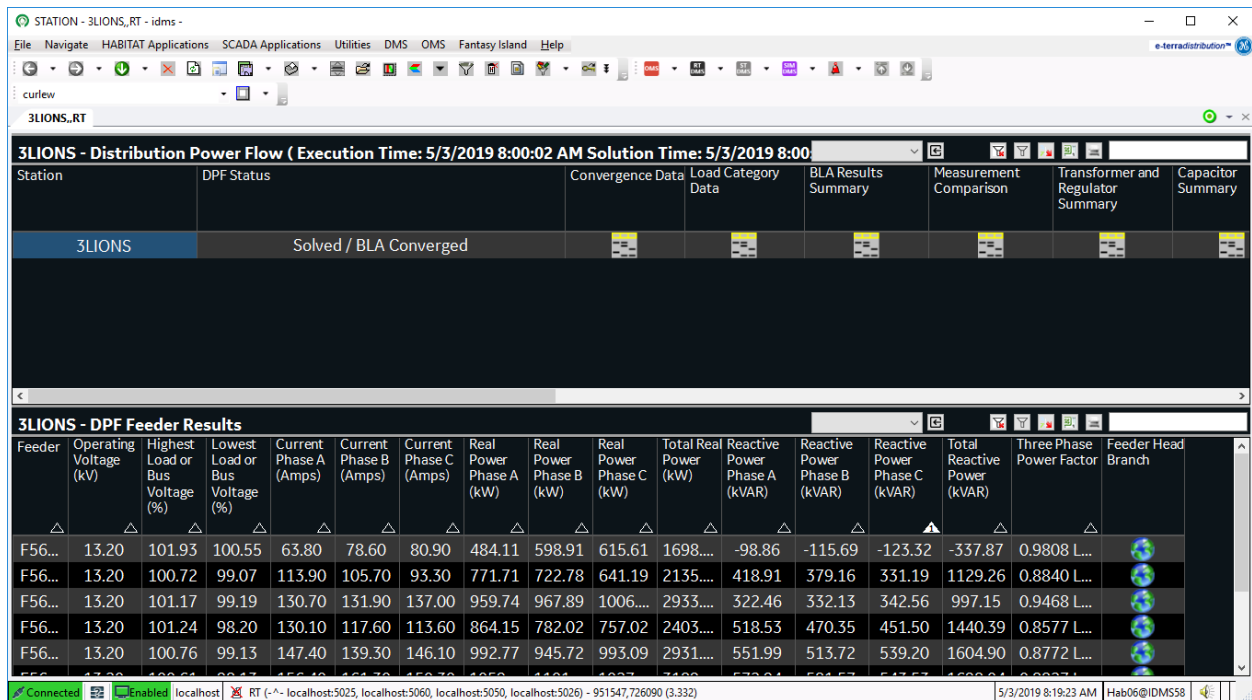


Figure 21. Example of a Multi-Pane Display

A multi-pane display is constructed by defining each of the panes as a separate grid tabular display, and then combining them using the frames capability of the ADMS Viewer.

The Netview command follows the standard format for calling up a grid tabular display. However, each of the displays is assigned to display frames in the argument of the command.

For example, the command to call up the Station Output display is:

```
NETVIEW://%DYN/GRIDTABULAR//%NAME?FRAME1=subDpf_griddisplay&FRAME2=FeederDpf_griddisplay
```

The two grid tabular display definition files are:

- SubDPF_griddisplay.xml
- FeederDPF_griddisplay.xml

Note

The .xml extension is not required in the above command.

3.4.4. Example 4: Multi-Select Display

The Multi-Select display allows you to configure the context menu for cases when several items or rows are selected in a grid tabular through the following methods:

- Using key combinations, such as Ctrl+Click and Shift+Click.
- Clicking check boxes on the Multi-Select column.
- Clicking the check box located in the header cell of the Multi-Select column. In this case, all rows are selected or deselected.

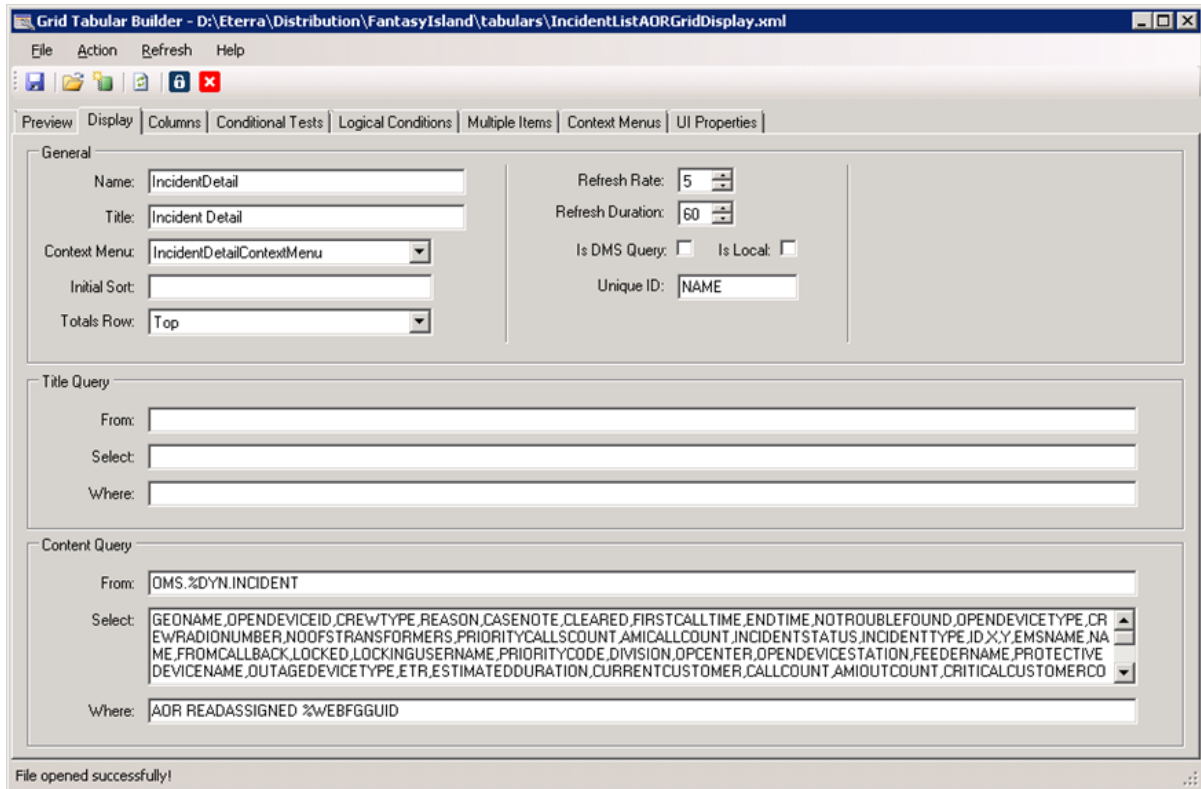
Context menu items are also processed depending on a row selection's state, as configured on the Grid Tabular Builder Context tab (see [Context Menus Tab](#)).

Incident Detail																
											Division	Op Center	Station	Feeder		
△	△				△	△	△	△				△		△		
												NORTH	WST	ROYAL TRO...	F5863	
												NORTH	WST	ROYAL TRO...	F5863	
												NORTH	WST	ROYAL TRO...	F5863	
												NORTH	WST	WINGED FO...	F7864	
												NORTH	WST	RAVEN	F8863	
												NORTH	WST	ROYAL TRO...	F5868	
												NORTH	WST	3LIONS	F5634	

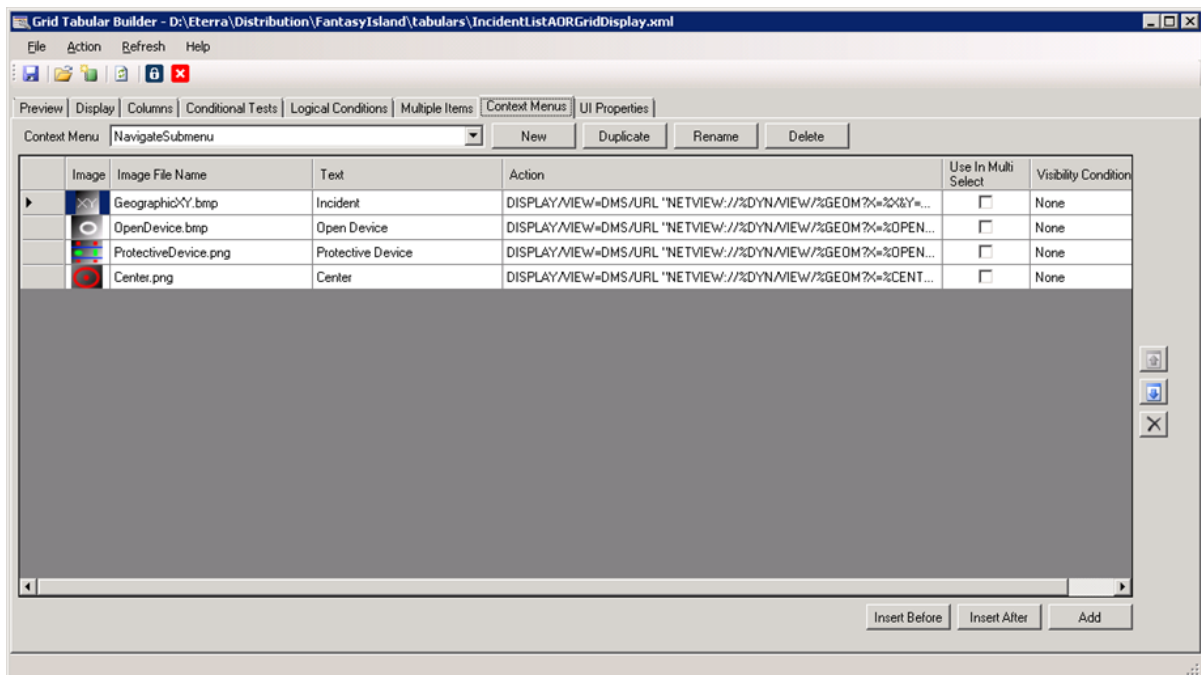
Figure 22. Multi-Select Feature Example

To add the multi-select feature to a grid tabular display:

1. Open the Grid Tabular Builder.
2. Open or create a grid tabular display.
3. Open the Display tab to define the Unique ID field. The Unique ID field contains a unique grid column name for data row identification purposes. This option is necessary to support sorting and filtering functions.



4. Open the Context Menus tab.
5. Select the check boxes in the "Use In Multi-Select" column to add the selected items to a context menu for multiple selected rows on the grid tabular display.



6. Open the Columns tab and click Add. The Add New Column dialog box appears.

7. In the Add New Column window:

- For the Placement Type field, select the CheckBoxMultiSelect option.

Note

Only one CheckBoxMultiSelect placement type is allowed.

- In the Name field, enter "Select" to ensure correct processing of actions for the OMS display.
 - In the Default Width field, enter a value larger than 20 to prevent overlapping the checkbox column with a neighboring column.
8. Click Add. The multi-select feature is configured and installed in the current grid display.
9. Save the grid display to the Tabulars folder. The default location of the Tabulars folder is D:\Eterra\Distribution\FantasyIsland\tabulars\.
10. To see the updated grid display, open, for example, the OMS display and enter the URL in the following format:

```
NETVIEW://RT/GRIDTABULAR/<__SAVED GRID DISPLAY NAME__>
```

Where *SAVED GRID DISPLAY NAME* is the grid display file name.

3.4.5. Example 5: Auto-Refreshable Display

To configure the refresh rate of a grid tabular display:

- On the Display tab, set the Refresh Rate and Refresh Duration fields (in seconds).

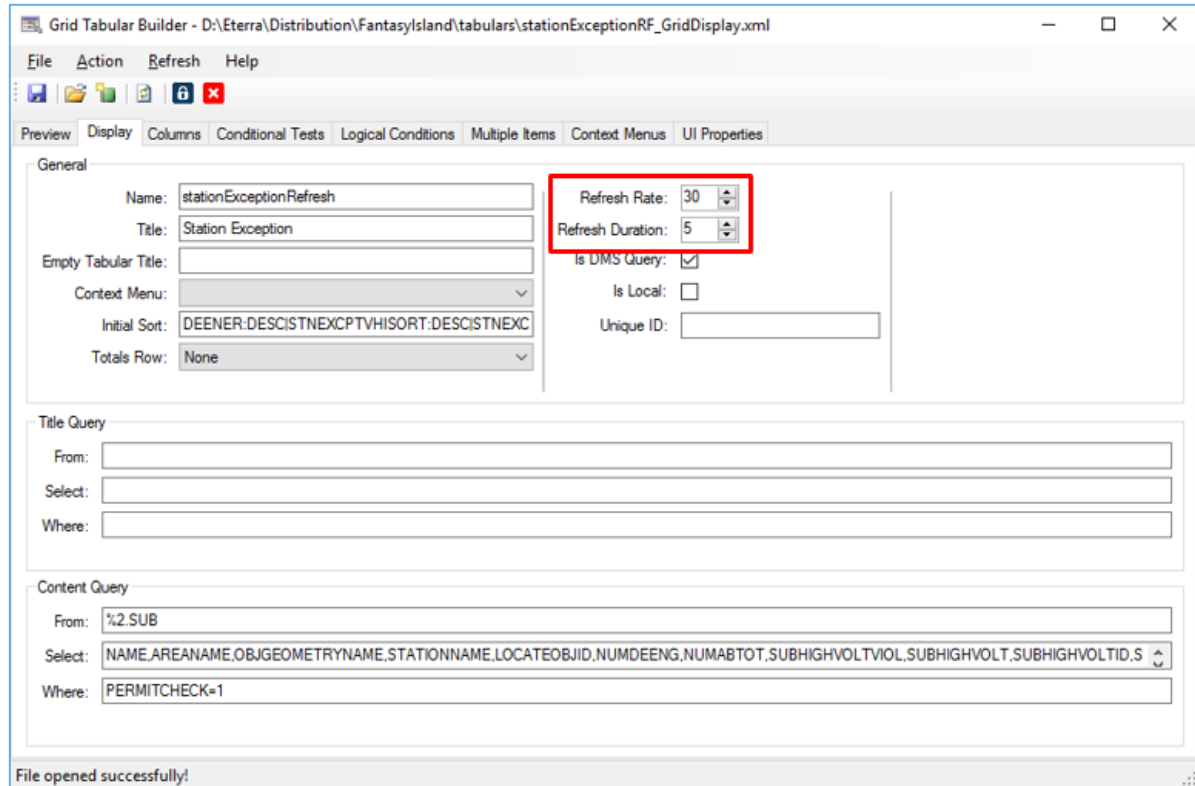


Figure 23. Auto-Refreshable Display Example

3.4.6. Example 6: Transformers Tree View Display

The Transformers Tree View display lists transformer objects in the system and includes an expandable hierarchical element tree.

To build the Transformers Tree View display:

1. In the Grid Tabular Builder, open the DeviceTreeView_GridDisplay.xml file and re-save it as TransformersTreeView_GridDisplay.xml.
2. Modify values on the Display tab as follows:
 - Name: TransformersTreeView_GridDisplay
 - Title: Transformers Tree View
 - Content Query From: RT.SUB.TF
 - Content Query Select:
DNOMTABNAME, ORGLEVEL_1_ID, TOPOSUBNAME, FEEDERNAME, ID, NAME, DEVICESERVCTRNAME, OBJGEOMETR

YNAME, LOCATEOBJID

The screenshot shows the 'Grid Tabular Builder' application window. The title bar indicates the file path: 'D:\Eterra\Distribution\FantasyIsland\tabulars\TransformersTreeView_GridDisplay.xml'. The menu bar includes 'File', 'Action', 'Refresh', and 'Help'. Below the menu is a toolbar with icons for file operations and a lock icon. The main area has several tabs: 'Preview', 'Display', 'Columns', 'Conditional Tests', 'Logical Conditions', 'Multiple Items', 'Context Menus', and 'UI Properties'. The 'Display' tab is active, showing a 'General' section with the following fields:

- Name: TransformersTreeView_GridDisplay
- Title: Transformers Tree View
- Empty Tabular Title: (empty)
- Context Menu: (dropdown menu)
- Initial Sort: NAME:ASC
- Totals Row: None
- Refresh Rate: 0
- Refresh Duration: 0
- Is DMS Query: ☒
- Is Local: ☐
- Unique ID: ID

Below the General section are two query sections:

- Title Query:** Fields for 'From:', 'Select:', and 'Where:' are present, but empty.
- Content Query:** Fields for 'From:', 'Select:', and 'Where:' are present. The 'From:' field contains 'RT.SUB.TF'. The 'Select:' field contains a list of fields: 'DNOMTABNAME, ORGLEVEL_1_ID, TOPOSUBNAME, FEEDERNAME, ID, NAME, DEVICSERVCNTRNAME, OBJGEOMETRYNAME, LOCATEOBJID'. The 'Where:' field is empty.

A status bar at the bottom of the window displays the message: 'File saved successfully!'.

3. Open the Columns tab.

4. Open the Level column for editing. Modify values as follows:

- Cell Name: %DNOMTABNAME | %ORGLEVEL_1_ID | %TOPOSUBNAME | %FEEDERNAME | %ID

This query specifies that the hierarchical element tree shows the DNOM Record Type, Service Center, Substation, Feeder, and Device ID nodes.

Note

The Level column placement type is "TreeView"

5. Open the Service Center column for editing. Modify values as follows:

- Header Cell Name: DEVICESERVCNTRNAME
- Cell Name: %DEVICESERVCNTRNAME

6. Remove the Is SCADA and Is DER columns.

7. Save the display file.

8. Within the ADMS client, execute the following command:

```
NETVIEW://RT/popup/DEVICETREEVIEW?TABULAR=TransformersTreeView&LEVEL_NAMES=DnomRecordType,ServiceCenter,Station,Feeder,Device
```

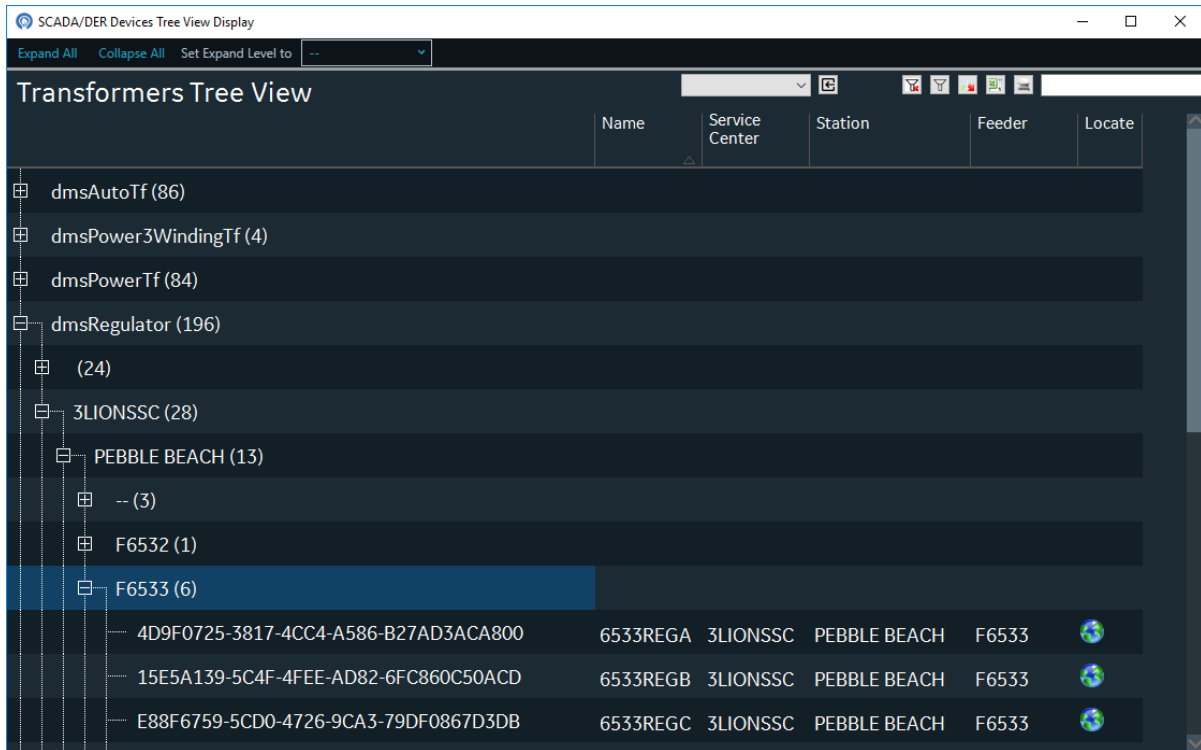
This command specifies that the TransformersTreeView display is opened in the Device Tree View popup, which is designed for hierarchy tree view displays. This command uses the following parameters:

- TABULAR: Specifies which grid tabular display opens in the Device Tree View popup.
-

LEVEL_NAMES: Defines option labels in the Set Expand Level To drop-down list on the Device Tree View popup.

- EXPANDTONODE: Optionally specifies which hierarchy levels appear expanded when the display is called.

9. Verify that the display looks similar to the figure below and works as expected.



3.4.7. Example 7: Display with a Row Index

This example shows how to configure row index columns to show the row index and the sorting index.

To configure a row index columns:

1. In the Grid Tabular Builder, open the IncidentListAORGridDisplay.xml file.
2. Open the Columns tab.
3. Add a column for the row index.

For example:

- Name: Row Index
- Header Cell Name: Row Index
- Placement Type:RowIndex

Add new column

Field

Name:

Header Cell Name:

Cell Name:

ToolTip:

Action Left:

Action Enter:

Context Menu:

Visibility:

Logical Col Visibility:

Total Type:

Type

Placement Type:

Precision Value:

Layout

Alignment:

Default Width: Minimum Width:

☐ Fix ☐ Hide ☐ Sortable ☐ Frozen

Visibility for Client Role

Excluded Roles: Included Roles:

4. Edit the column that shows the number of de-energized customers (# Cust).

- Header Cell Name: CustOutCount

This allows using the CustOutCount name for sorting index column sorting criteria.

5. Add a column for the sorting index.

For example:

- Name: Sorting Index by Affected Customers
- Header Cell Name: Sorting Index by Affected Customers
- Cell Name: CustOutCount:DESC

This sorting value specifies that the sorting index is calculated for the number of de-energized customers in the incident, in the descending order. This sorting index may be used to show incident priority or severity based on the number of affected customers.

- Placement Type: RowIndex

Field

Name:

Header Cell Name:

Cell Name:

ToolTip:

Action Left:

Action Enter:

Context Menu:

Visibility:

Logical Col Visibility:

ETRPAST

ETREXPIRINGSOON

ETRCLOSE

ITRPAST

ITREXPIRINGSOON

Total Type:

Type

Placement Type:

Precision Value:

Layout

Alignment:

Default Width:
Minimum Width:

☐ Fix
☐ Hide
☐ Sortable
☐ Frozen

Visibility for Client Role

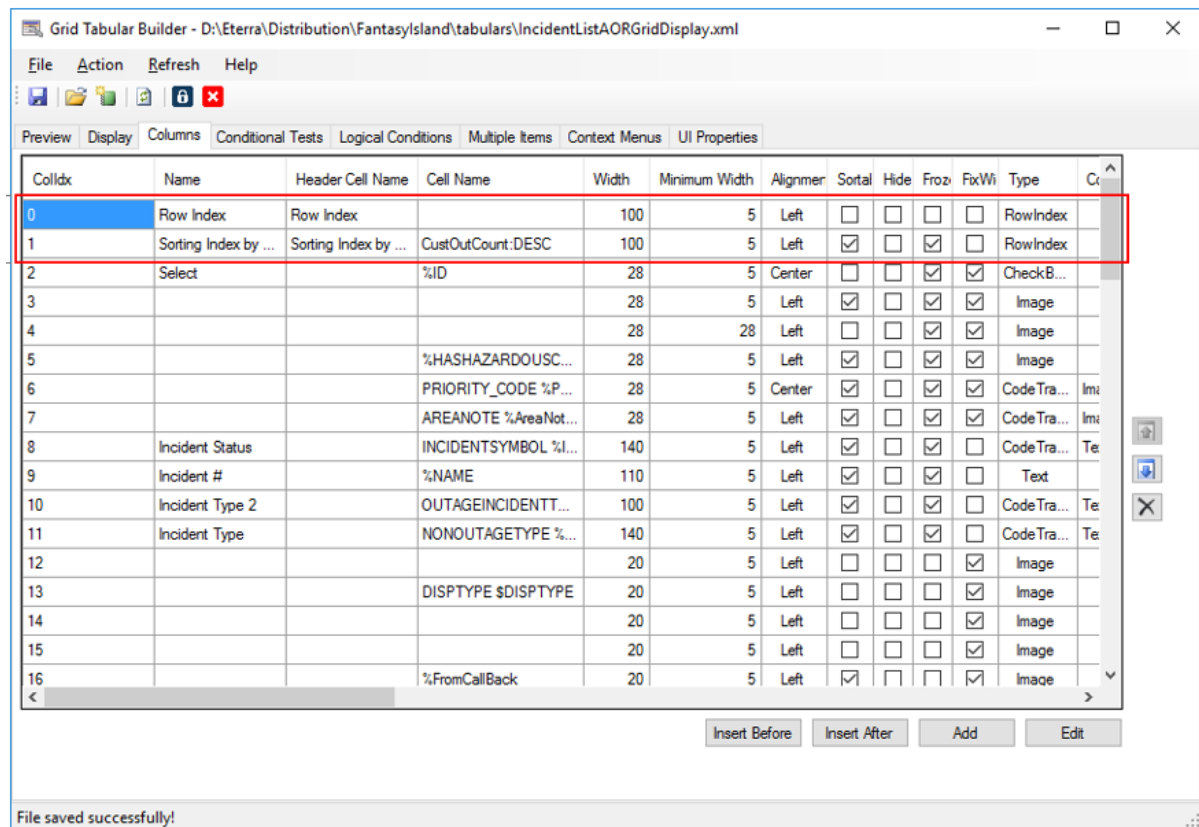
Excluded Roles

Included Roles

Add

Cancel

6. Make sure that the columns are listed in the Columns tab.



- Save the display file.
- Within the ADMS client, create several incidents and open the Incidents tab of the OMS display plugin.
- Verify that the display shows the row index and the sorting index based on the number of de-energized customers in the incident.

Overview

Incidents (6)

Reset Filter

General Electric

Incident Detail

Row Index	Sorting Index by Affected Customers					Incident Status	Incident #	Incident Type 2	Incident Type	# Cust	# Calls
										84	1
1	2					New	0630000006		Outage	34	0
2	1					New	0630000005		Outage	38	0
3	3					New	0630000004		Outage	10	0
4	5					New	0630000003		Outage	1	0
5	6					IS Incident	0630000002		Hazard	0	1
6	4					New	0630000001		Outage	1	0

4. Standard Grid Tabular Displays

This chapter includes examples of the key grid tabular displays that are included in the standard ADMS product, with comments on particular types of data presentation.

For descriptions of all grid tabular displays, refer to the *ADMS Display Reference Manual*.

4.1. Station Exception

The Station Exception display summarizes abnormalities and violations by station, and is used as a basis for navigating to more-detailed information about the exceptions. A separate Station Exception is provided for real-time, study, and simulator modes.

Display call-up URL:

```
netview://RT/gridtabular/stationException_GridDisplay
```

The Station Exception provides a high-level perspective of the activities within a dispatcher's Area of Responsibilities (AOR). It only displays stations where the dispatcher has been granted write access.

By default, substations are sorted and presented to the user to show the worst conditions in the following priority order:

1. De-energization (number of de-energized customers).
2. High-voltage violations.
3. Low-voltage violations.
4. Voltage imbalance violations.
5. Overload violations.
6. Flow imbalance violations.
7. Ground return current rate.
8. DPF solution status.
9. Number of abnormal topology conditions.

For voltage and flow conditions, the worst result for each station is shown. If the worst result represents a violation, it is highlighted in red. The violation values are linked to displays giving more details.

Hidden columns that drive the display's default sorting order are an important feature of this display.

Selecting a substation name navigates to an exception summary for each feeder of that substation.

4.2. De-Energized Customers

The De-Energized Customers display shows de-energized loads and transformers along with the associated customer count and the date and time of de-energization. This display can be accessed via the station exception or the feeder summary display.

Display call-up URL:

```
netview://RT/gridtabular/deenergization_GridDisplay/<Substation Name>
```

Clicking the Locate button navigates to the specified device on the geographic display.

4.3. Station Abnormal Summary

The Station Abnormal Summary display shows switches and permanent jumpers in abnormal positions, abnormally energized loads, temporary grounds, cuts, and jumpers. This display can be accessed via the station or feeder summary.

Display call-up URL:

```
netview://RT/gridtabular/Abnormal_griddisplay/<Substation Name>
```

The date and time when the abnormal condition was detected are provided for switches, permanent jumpers in abnormal positions, temporary grounds, cuts, and jumpers. Each abnormal condition also provides a reference to the area, substation, and type of change that occurred.

Switches are listed by switch name, while temporary modifications have an internally generated name that includes the originating branch name and a distance.

Clicking the Locate button navigates to the specified device on the geographic display.

The Tag column indicates whether any tags have been applied to the listed components. A gray symbol indicates that no tag has been applied.

4.4. Measurement Data

The Measurement Data display provides a list of SCADA measurements known by ADMS for the selected station. In the real-time environment, these measurements are used as inputs to the DPF solution.

Display call-up URL:

```
netview://RT/gridtabular/measurements_GridDisplay/<Substation Name>
```

4.5. Station Output Summary

The Station Output Summary display provides navigation to DNAF results for a particular station. For example, Distribution Power Flow results for a station can be accessed from this display. This display can be easily tailored to remove or add the desired navigation points for your dispatcher.

Display call-up URL:

```
netview://RT/gridtabular/stnOutput_GridDisplay
```

4.6. DPF Station Results

The DPF Station Results display provides high-level DPF output for the substation. This display can be accessed from a button on the station output summary. It contains the following information:

- Information about DPF convergence for the station
- Value of supply bus voltage in KV, per unit
- Power transformer LTC tap positions and loadings
- Substation capacitor steps
- Voltage regulator tap positions and loadings
- Feeder capacitor status (ON/OFF)

Display call-up URL:

```
netview://RT/gridtabular/subDPF_GridDisplay/<Substation Name>/
```

4.7. Feeder Losses and Power Quality Indices

The Feeder Losses and Power Quality Indices display informs about losses related to elements along the feeder. This display can be accessed by clicking the Loss/PQI Summary button on the DPF Station Results display.

The information on this display is recalculated each time the DPF solves for a station. Instantaneous losses and Power Quality Index (PQI) analysis values are presented. Losses and PQI results for individual components can be accessed from the geographic display (operator output display).

Display call-up URL:

```
netview://RT/gridtabular/FeederLossesPQI_GridDisplay/<Substation Name>
```

4.8. Protection Validation

The Protection Validation display can be accessed from the Station Output Summary display. Protection Validation results are shown for all of the breakers and over current relays within a substation.

Display call-up URL:

```
netview://RT/gridtabular/faultanalysis_GridDisplay/<Substation Name>/
```

4.9. Mobile Substation List

The Mobile Substation List display lists all the mobile substations that are currently connected to the distribution network and contains key information related to these stations.

Display call-up URL:

```
netview://RT/gridtabular/MobileSubStationsGridDisplay/
```

4.10. Tag Summary

The Tag Summary display lists all the tags currently applied to the distribution network.

Display call-up URL:

```
netview://RT/gridtabular/TagSummary_GridDisplay/
```

4.11. Annotation Summary

The Annotation Summary display shows a list of the annotation types that can be applied to the distribution network. Selecting any of the annotation types navigates to a summary of the currently applied annotations of that type.

Display call-up URL:

```
netview://RT/gridtabular/AnnoSummary_griddisplay/<Annotation_Type>
```

4.12. Abnormal Summary

The Abnormal Summary display shows a list of the temporary modification types that can be applied to the distribution network. Selecting any of the modification types navigates to a summary of the currently applied modifications of that type.

Display call-up URL:

```
netview://RT/gridtabular/summaryCutJumpGrd_GridDisplay
```

4.13. Base Application Status

The Base Application Status display provides an interface where the dispatcher can view the status of base DNAF functions (DPF, FL and PRV) for each substation.

Display call-up URL:

```
netview://RT/gridtabular/BaseApplicationStatus_GridDisplay
```

4.14. Advanced Application Status

The Advanced Application Status tabular display shows the status of the network optimization functions (LVM, FISR, AFR, FISRCA, and PLOS) for each substation.

Display call-up URL:

```
netview://RT/gridtabular/dsocontrol_GridDisplay
```

4.15. Application Input Summary

The Application Input Summary (previously - DOAN Control) display provides the summary for inputs and measurements used for DNAF applications.

Display call-up URL:

```
netview://RT/gridtabular/ApplicationInputSummary_GridDisplay/<Substation Name>/
```

4.16. BLA Station Summary

For each substation, the BLA Station Summary display provides Bus Load Allocation (BLA) calculations by measurement island.

Display call-up URL:

```
netview://RT/gridtabular/BLAStationSummary_GridDisplay/<Substation Name>
```

4.17. LVM Station Summary

For each substation, the LVM Station Summary display provides LVM application results by station.

Display call-up URL:

```
netview://RT/gridtabular/LVMStationSummary_GridDisplay/<Substation Name>
```

4.18. Stations Equipment Commissioning and Decommissioning

The Stations Equipment Commissioning and Decommissioning display is accessible from the RT menu. The display shows a summary of whether each station currently contains equipment that is modeled in the distribution electrical network but not active (planned to be put in service, planned to be removed, or was recently removed).

Display call-up URL:

```
netview://RT/gridtabular/EquipmentCommissioningDecommissioning_GridDisplay
```

4.19. Pending Switches

The Pending Switches display lists all switches that currently have "pending permanent" status. This means that the normal state of the switch has been changed, but the change has not yet been confirmed by an update of the geographic information system/asset management system data. The network topology processing continues to use the existing normal state of the switch until the change is confirmed.

Display call-up URL:

```
netview://RT/gridtabular/pendingSwitches_GridDisplay/
```

Clicking the Locate button navigates to the specified device on the geographic display.

4.20. Find Results Summary

The Find Results Summary display is the tabular grid in the lower half of the Tabbed Find Device pop-up display. The Results Summary grid uses the Coded Translation functions to determine the correct bitmap and navigation command to display according to the device type listed in the table.

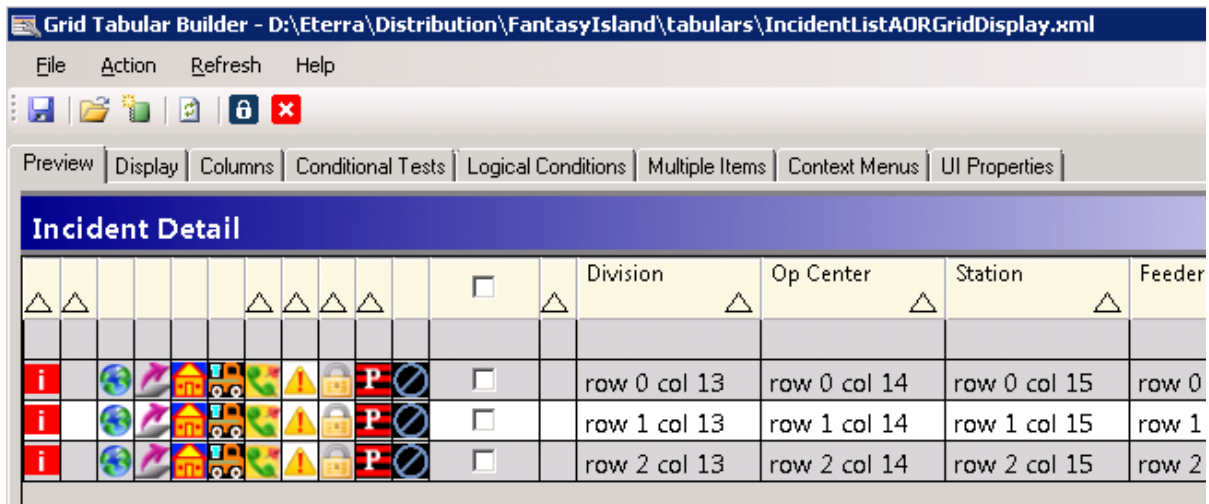
The display is defined by the FindResultsGridDisplay.xml file.

5. Using Coded Translation Items

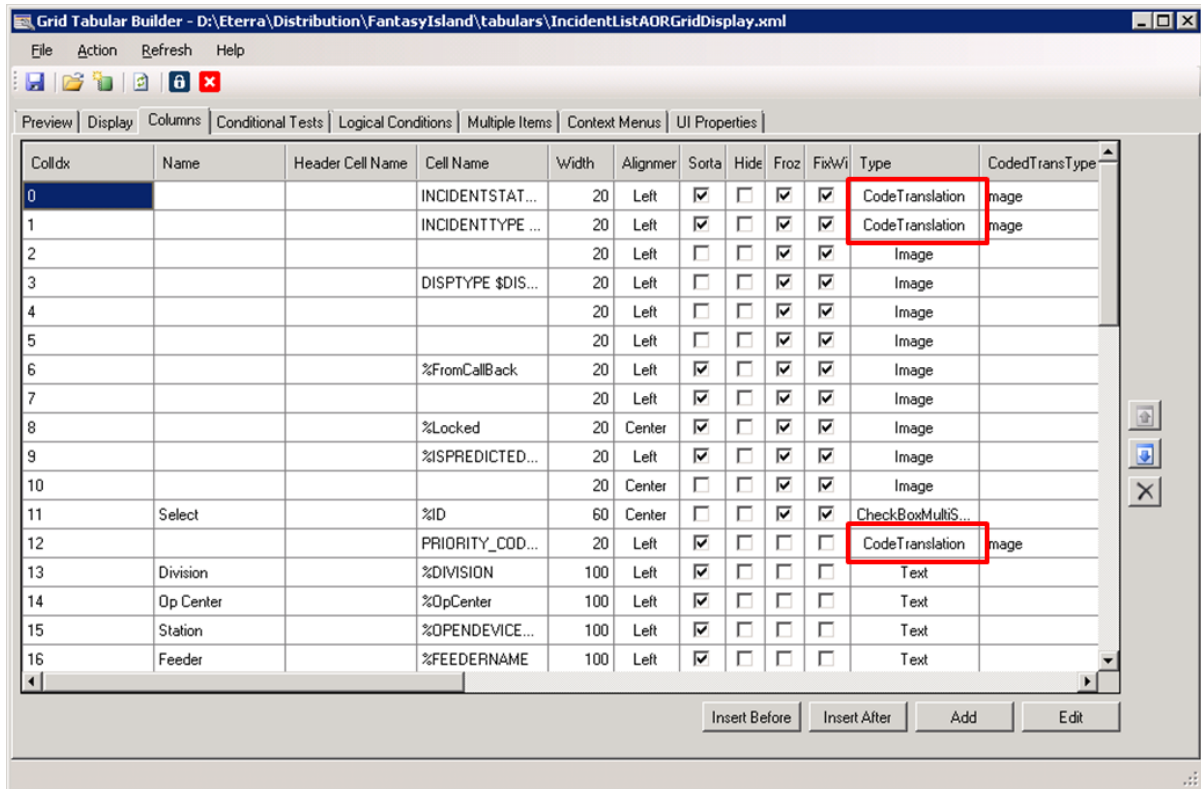
Each of ADMS tabular displays can be configured to use coded translation items (refer to the *Coded Translation Editor User Guide*). This allows displaying multiple bitmaps and text values that are assigned to each key of the coded translation item.

The example below shows how the Incident Detail display uses the IncidentType coded translation item.

1. Open the Grid Tabular Builder.
2. Open the IncidentListAORGridDisplay.xml file that configures the Incident Detail display.



3. On the Columns Tab, notice columns that use coded translation types.



4. Select the INCIDENTTYPE row (second), which contains the IncidentType coded translation item, and click Edit.

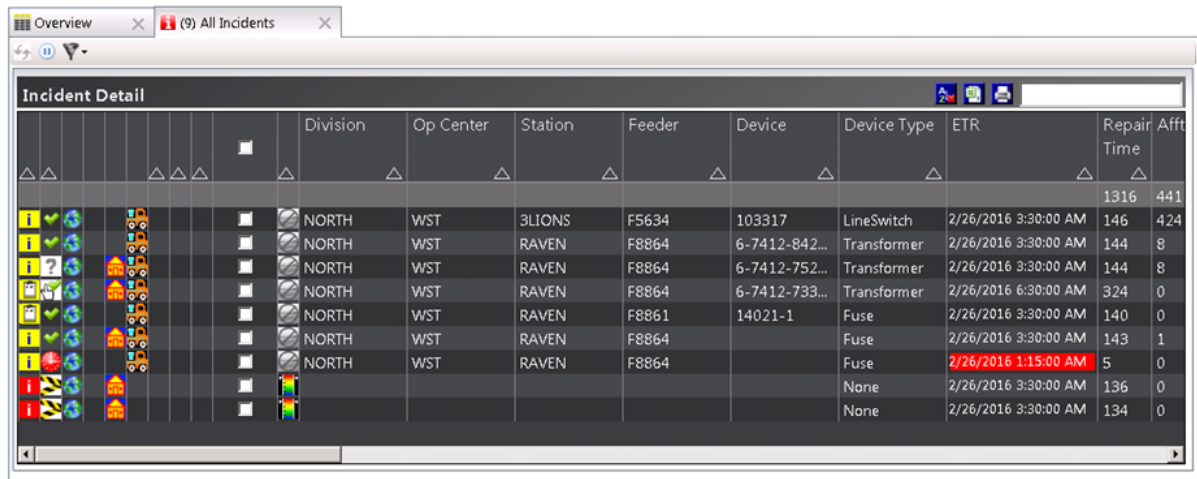
Each coded translation type is configured as follows:

- The Cell Name value format is *<Coded Translation Item> %<Query Helper>* (for example, INCIDENTTYPE %INCIDENTTYPE).
- The CodeTranslation placement type is specified in the Type group.
- The Image or Text option is selected to display an image or text.

Depending on the numeric value received, the INCIDENTTYPE column will show different incident type icons.

5. In the ADMS client, navigate to the Incident Details display.

Notice that the second column shows different images for different incident types.



The screenshot shows the 'Incident Detail' window in the ADMS client. The window has a title bar with 'Overview' and '(9) All Incidents'. Below the title bar is a toolbar with icons for search, refresh, and print. The main area is a table with the following columns: Division, Op Center, Station, Feeder, Device, Device Type, ETR, Repair Time, and Afft. The table contains 10 rows of incident data. The first column of the table (which is the second column of the window) contains various icons representing different incident types, such as a person, a truck, a fire, and a lightning bolt. The 'ETR' column shows dates and times, with one entry highlighted in red: '2/26/2016 1:15:00 AM'.

	Division	Op Center	Station	Feeder	Device	Device Type	ETR	Repair Time	Afft
								1316	441
	NORTH	WST	3LIONS	F5634	103317	LineSwitch	2/26/2016 3:30:00 AM	146	424
	NORTH	WST	RAVEN	F8864	6-7412-842...	Transformer	2/26/2016 3:30:00 AM	144	8
	NORTH	WST	RAVEN	F8864	6-7412-752...	Transformer	2/26/2016 3:30:00 AM	144	8
	NORTH	WST	RAVEN	F8864	6-7412-733...	Transformer	2/26/2016 6:30:00 AM	324	0
	NORTH	WST	RAVEN	F8861	14021-1	Fuse	2/26/2016 3:30:00 AM	140	0
	NORTH	WST	RAVEN	F8864		Fuse	2/26/2016 3:30:00 AM	143	1
	NORTH	WST	RAVEN	F8864		Fuse	2/26/2016 1:15:00 AM	5	0
						None	2/26/2016 3:30:00 AM	136	0
						None	2/26/2016 3:30:00 AM	134	0

6. Action URLs

This chapter describes the action URLs that specify the action to be taken when the user clicks an item associated with that action. This applies to the Select Action Left and Select Action Enter columns on the Columns tab and to the Action column on the Context Menus tab.

An action URL command string uses the following syntax:

```
NETVIEW://%DYN/command1/command2/command3/command4?parameter1&parameter2
```

This syntax is explained below:

- **%DYN:** The name of dynamics (for example, RT, ST, and SIM).
- **Command1:** The command (for example, VIEW, GRIDTABULAR, NAVOV, and UPDATE).
- **Command2:** Geometry name, a component type, or a display name, depending on the content of command1.
- **Command3:** A station name or a file name, depending on the content of command1.
- **Command4:** The action, component name, or component ID.
- **Parameter:** Additional data associated with each command. Parameters are separated with an "&" symbol.

To open a new network view viewport for an action URL, use the VIEWNEW command. Alternatively, you can use one of the following prefixes. Note that when a prefix is used, an action URL must be enclosed in quotes.

- DISPLAY/VIEW=DMS/URL
- DISPLAY/URL
- FGCMDDISPLAY/VIEWPORT=SWITCHORDER/URL
- FGCMDDISPLAY/URL

For more information about Netview commands, refer to the *Viewer Configuration Editor User Guide*.

6.1. Navigation URLs

To navigate to a specific location on the geographic network view, use a URL such as the following:

```
NETVIEW://%DYN/VIEW/%GEOM?X=%X&Y=%Y&ZOOM=3.9&FORCENONMAINON&HIDELABELS&CustomerLayers=(SHOWACTIVECUST)
```

All variable names (prefixed with %) are replaced by their associated values before the URL is parsed, so these variables must exist in the Select clause of the Content Query of the grid tabular display. In the example above, if the Content Query does not select variables X and Y, the network view is unable to navigate to a specific point.

Examples of navigation URLs

- The following URL navigates to the location specified by the given X and Y coordinates, such as for an incident:

```
DISPLAY/VIEW=DMS/URL  
"NETVIEW://%DYN/VIEW/%GEOM?X=%X&Y=%Y&ZOOM=3.9&FORCENONMAINON&HIDELABELS&CustomerLayers=(SHOWACTIVECUST)"
```

- The following URL navigates to the location of the open device associated with an incident:

```
DISPLAY/VIEW=DMS/URL  
"NETVIEW://%DYN/VIEW/%GEOM?X=%OPENDEVICEPTX&Y=%OPENDEVICEPTY&ZOOM=3.9&FORCENONMAINON&HIDELABELS&CustomerLayers=(SHOWACTIVECUST)"
```

- The following URL navigates to the center of the network view:

```
DISPLAY/VIEW=DMS/URL  
"NETVIEW://%DYN/VIEW/%GEOM?X=%CENTERPTX&Y=%CENTERPTY&ZOOM=3.9&FORCENONMAINON&HIDELABELS&CustomerLayers=(SHOWACTIVECUST)"
```

- The following URL navigates to an assigned crew:

```
NETVIEW://%DYN/VIEW/%GEONAME?X=%CURRENTLOCATIONX&Y=%CURRENTLOCATIONY&
```

- The following URL navigates to a device:

```
NETVIEW://%DYN/VIEW/%OBJGEOMETRYNAME/%STATIONNAME?LOCATE=%ID&FORCENONMAINON&ZOOM=20&ADJACENT=2
```

- The following URL navigates to a station internals view:

```
NETVIEW://%DYN/VIEW/%EMSNAME/%EMSNAME
```

- The following URLs navigate to a container internals view:

```
NETVIEW://%DYN/VIEWNEW/Internal-  
%ID/%NELECSUBNAME?SHOWLABELS&ZOOM=100&APPENDDISPLAYTITLE=%SWCABTYPE,%ADDRESS.NOAPPSTR  
NETVIEW://%DYN/SCHEMATICNEW/CONTAINER/%UICONTSTATNAME/%ID?SHOWLABELS&COLORBYVOLTAGE&ZOOM=5  
&FORCENONMAINON&%ADDRESS.NOAPPSTR
```

- The following URL navigates to a substation maximum loading segment:

```
NETVIEW://%DYN/GEO/%GEOM/%PLANMAXLOADSEGSUBID?LOCATE=%PLANMAXLOADSEGIDONLY
```

6.2. Grid Tabular URLs

- To load a DMS grid tabular display, enter a URL such as the following:

```
NETVIEW://%DYN/GRIDTABULAR/deenergization_griddisplay/%NAME
```

- To load several DMS grid tabular displays in separate frames (up to 4 are supported), enter a URL such as the following:

```
NETVIEW://%DYN/gridtabular://%NAME?Frame1=StationCapInput1_GridDisplay&Frame2=StationCapInput2_GridDisplay&Frame3=StationCapInput3_GridDisplay
```

- To load an OMS grid tabular display, enter a URL such as the following:

```
NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/INCIDENTS_AOR
```

- To load a specific display for the selected OMS component (such as a division, incident, or crew), enter a relative path such as the following:

```
SUB?DISPLAY=NOCUSTOMERLISTGRIDDISPLAY&7=%NAME
```

In the example above, NAME refers to the unique ID (such as of an incident) that is used to select the customer list.

Examples of Grid Tabular URLs

- The following URL invokes the feeder exception summary display for the selected substation:

```
NETVIEW://%DYN/gridtabular/feederexception_griddisplay/%NAME
```

In the example above, NAME refers to the unique ID (such as, from an incident) that is used to select the customer list.

- The following URL invokes two grid tabular displays in separate frames:

```
NETVIEW://%DYN/GRIDTABULAR://%NAME?FRAME1=subDpf_griddisplay&FRAME2=FeederDpf_griddisplay
```

In the example above, NAME refers to the unique ID of a substation.

- The following URL calls the display that contains the list of employees:

```
NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/EMPLOYEES
```

- The following URL calls the display that lists the crews assigned to an incident:

```
NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/INCIDENT?ID=%ID&TABNAME=crewsTabPage
```

In the example above, ID refers to the unique ID of an incident.

- The following URL calls the display that shows the Op Center Summary display for the selected division:

```
SUB?DISPLAY=OPCENTERSUMMARYFORGRIDDISPLAY&7=PARENTNAME&8=%NAME
```

In the example above, Name refers to the unique name of a division.

6.3. Pop-Up URLs

Pop-ups are dialog boxes that display data and perform functions for a specific row of a grid tabular display.

6.3.1. DMS Pop-Up URLs

To display a DMS pop-up, enter a URL such as the following:

```
NETVIEW://%DYN/POPUP/_<popup_name>_
```

Where:

- *<popup_name>* is one of the following DMS pop-up names:
 - CUT
 - TEMPGROUND
 - FAULT
 - JUMPER
 - CUSTOMERCORRECT
 - DNAFSOLUTIONS
 - SAVECASE
 - FIND
 - TAG
 - ANNOTATION
 - ANNOTATIONUPDATE
 - MOBILESUB
 - SPECIALANNOTATION
 - SPECIALANNOTATIONUPDATE
 - SWITCHSTATUS
 - CAPACITORSTATUS
 - BLOCKPREDICTIONSTATUS

- CURTAILPLAN
- SCHEDULEPOPOP
- SWOQualifications
- SWOEmployee

Examples of DMS pop-up URLs

- To call a Schedules pop-up display for the selected distribution transformer:

```
NETVIEW://%DYN/POPOP/SCHEDULEPOPOP/%TOPOSUBNAME/%ID?TYPE=DYNAMIC&OBJECTTYPE=%DNOMOBJECTTYPE&am
p;SCHEDULETYPES=1,7
```

- To call an Annotation pop-up to edit an annotation:

```
NETVIEW://%DYN/POPOP/ANNOTATIONUPDATE/%ID
```

- To call a Curtailment pop-up for the selected feeder:

```
netview://%DYN/POPOP/Curtailplan/%TOPOSUBNAME?FilterFeeder=%DYNFEEDERNAME
```

- To call a DNAF Solution pop-up for the selected component:

```
NETVIEW://%DYN/POPOP/DNAFSOLUTIONS/%DYNFEEDERSUBNAME/%ID
```

6.3.2. OMS Pop-Up URLs

To display an OMS pop-up, enter a URL such as the following:

```
NETVIEW://%DYN/<module>/<popup_name>
```

Where:

- *<module>* is the plugin module (ADDIN, NETADDIN, or POPUP) that loads the pop-up. The ADDIN and NETADDIN modules indicate that the pop-up is a dynamically loaded display.
- *<popup_name>* is the pop-up name. The following is a list of existing OMS pop-ups.
 - INCIDENTFORM
 - CREWUPDATEFORM
 - PLANNEDOUTAGE
 - BLOCKPREDICTIONSTATUS
 - CUSTOMERCORRECT
 - ADDCALLPLUGINFORM
 - ADDUNASSOCIATEDCALLPLUGINFORM

Examples of OMS pop-up URLs

- To call an Update Planned Outage pop-up for the selected device:

```
NETVIEW://%DYN/ADDIN/PLANNEDOUTAGE?ID=%ID&STATION=%STATION&DEVICENAME=%DEVICENAME
```

- To call a Merge Incidents pop-up display:

```
NETVIEW://%PLUGIN.%DYN/NETADDINS/INCIDENTFORM?TYPE=INCIDENTLIST&DISPLAYTYPE=MERGEINCIDENT&ID=%ID&INCIDENTIDS=%ID
```

- To call a Crew Update pop-up display:

```
NETVIEW://%PLUGIN.%DYN/NETADDINS/CREWUPDATEFORM?ID=%ID
```

- To call an Associate with Incident pop-up display:

```
NETVIEW://%PLUGIN.%DYN/NETADDINS/INCIDENTFORM?TYPE=INCIDENTLIST&DISPLAYTYPE=ASSOCIATEINCIDENT&ID=%SELECTCOMPID
```

- To call a Block Prediction pop-up display:

```
NETVIEW://%DYN/POPOP/BLOCKPREDICTIONSTATUS/%TOPOSUBNAME/%ID
```

- To call a Customer Correction pop-up display:

```
NETVIEW://%DYN/POPOP/CUSTOMERCORRECT?ID=%ID&NAME=%NAME&FEEDERNAME=%DYNFEEDERNAME&SERVICEPOINTID=%SERVICEPOINTID
```

- To call an Add/Update Customer Call pop-up display:

```
NETVIEW://%PLUGIN.%DYN/OMSDISPLAYPLUGIN/ADDCALLPLUGINFORM?ID=%ACCOUNTNUMBER
```

- To call an Add/Update Unassociated Call pop-up display:

```
NETVIEW://%PLUGIN.%DYN/OMSDISPLAYPLUGIN/ADDUNASSOCIATEDCALLPLUGINFORM?X=%X&Y=%Y&ID=%ID
```

6.4. Message Box URLs

A "message box" is a standard Microsoft Windows dialog box that displays a text message. Message boxes are useful for showing multi-line text fields (for example, comments) that do not fit well in a grid tabular display.

To display a message box, enter a URL such as the following:

```
MessageBox?text=<__text__>&caption=<__caption__>
```

Where:

- `<text>` is the text to display in the message box.
- `<caption>` is the title of the message box. If a caption is not specified, the column name that was clicked is used as the caption.

Examples of Message Box URLs

Edit existing column

Field

Name:

Header Cell Name:

Cell Name:

ToolTip:

Action Left:

Action Enter:

Context Menu:

Visibility:

Logical Col Visibility:

Total Type:

Type

Placement Type:

Precision Value:

Image:

Image:

Layout

Alignment:

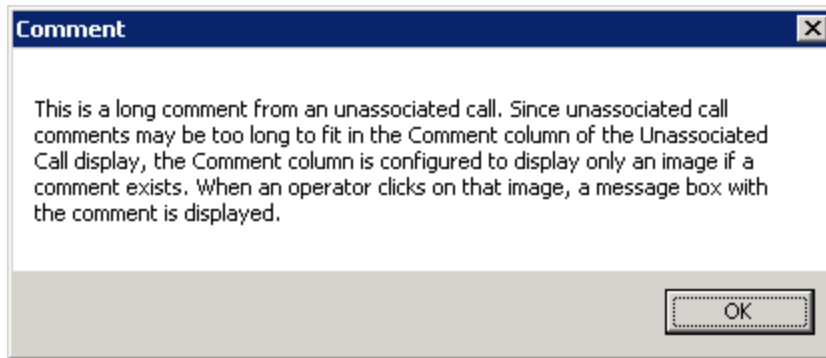
Default Width:

☐ Fix ☐ Hide ☒ Sortable ☐ Frozen

Example 1

`MessageBox?text=%COMMENT`

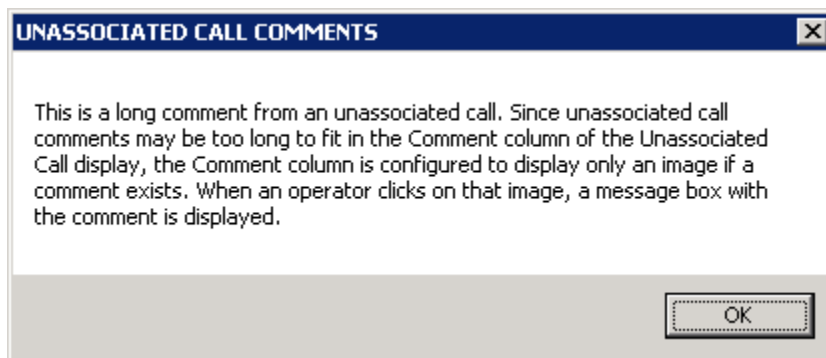
displays the value of `%COMMENT` in a message box whose caption is the column name, as shown below.



Example 2

```
MessageBox?text=%COMMENT&caption=Unassociated Call Comments
```

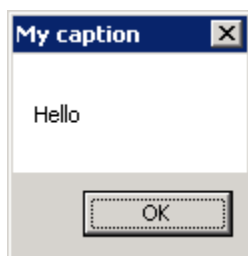
displays the value of %COMMENT in a message box whose caption is "Unassociated Call Comments", as shown below.



Example 3

```
MessageBox?text=Hello&caption=My Caption
```

displays "Hello" in a message box whose caption is "My Caption", as shown below.



6.5. DNAF Manual Entry Display URLs

Manual entry display URLs use the following syntax:

```
NETVIEW://%DYN/DNAFPARAMS/<display_name>/<station_name>/<unique_id>
```

Where:

- *<display_name>* is the manual entry display name. The following is a list of existing manual entry displays:
 - BUSPARAMS
 - CAPACITORPARAMS
 - DNAFCONFIG
 - DNAFAGGREGATION
 - ENERGYRESPARAMS
 - GENERATORPARAMS
 - INVERTERPARAMS
 - LINEPARAMS
 - LOADPARAMS
 - MANUALPARAMS
 - REACTORPARAMS
 - STNINPUTNAME
 - SWITCHPARAMS
 - TFPARAMS
- *<station_name>* is the name of the substation the device belongs to.
- *<unique_id>* is the identifier that uniquely identifies a component that the command applies to.

Examples of DNAF manual entry display URLs

- To call a Bus Parameters display:

```
NETVIEW://%DYN/DNAFPARAMS/BUSPARAMS/%TOPOSUBNAME/%ID
```

- To call a Manual Entry Parameters display:

```
NETVIEW://%DYN/DNAFPARAMS/MANUALPARAMS
```

- To call a Manual Entry Parameters display:

```
NETVIEW://%DYN/DNAFPARAMS/DNAFAGGREGATION
```

6.6. SCADA Display URLs

- To call a SCADA Online display, enter a URL such as the following:

```
DISPLAY/APP=SCADA/VIEWPORT=%EMSNAME %EMSNAME  
FGCMD DISPLAY/VIEWPORT=%NAME/APP=SCADA %NAME
```

- To call a SCADA Control display, enter a URL such as the following:

```
NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%MEASNAMEFOROBJECT/SCADA?PICTURE=fantasyisland_internals_scada_  
control_popup.fgpopup&LOCATION=1&FOLLOWMONITOR=1
```

- To call a Tagging Operations display, enter a URL such as the following:

```
NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%MEASNAMEFOROBJECT/SCADA?PICTURE=FANTASYISLAND_INTERNALS_SCADA_  
CONTROL_POPUP.FGPOPUP&LOCATION=1
```

- To call a SCADA Switch Popup display, enter a URL such as the following:

```
NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%MEASNAMEFOROBJECT/SCADA?PICTURE=fantasyisland_internals_scada_  
switch_popup.fgpopup&LOCATION=1
```

6.7. Attribute Panel URLs

The URL syntax to call an attribute panel is as follows:

```
NETVIEW://%DYN/ATTRIBUTES/_<panel_type>/_<__unique_id__>
```

Where:

- *<panel_type>* is the panel type, such as the following:
 - ANALYSTATTRIBUTE
 - ANALYSTOUTPUTS
 - OPERATORATTRIBUTE
 - OPERATOROUTPUTS
- *<unique_id>* is the identifier that uniquely identifies an instance that the command applies to. It may include several identifiers.

Examples of Attribute Panel URLs

- To call an Operator Attributes panel for a component, enter a URL such as the following:

```
NETVIEW://%DYN/ATTRIBUTES/OPERATORATTRIBUTE/%DNOMTABNAME://%TOPOSUBNAME/%ID
```

- To call an Analyst Output panel for a component, enter a URL such as the following:

```
NETVIEW://%DYN/ATTRIBUTES/ANALYSTOUTPUT/%DNOMOBJECTTYPE://%TOPOSUBNAME/%ID
```

6.8. Control URLs

The URL syntax for control URLs is as follows:

```
NETVIEW://%DYN/ControlCommand?parameter1&parameter2
```

Examples of Control URLs

- Enter URLs such as the following:

```
NETVIEW://%DYN/REMOVEPLAN?ID=%ID&STATION=%PLANSTATIONNAME
NETVIEW://%DYN/CANCELPLAN?ID=%ID&STATION=%PLANSTATIONNAME
NETVIEW://%DYN/EXECUTEPLAN/PLAN/%OPTPLANSUBNAME/%PLANFUNCID/IMPLEMENT
NETVIEW://%DYN/EXECUTEPLAN/PLAN/%OPTPLANSUBNAME/%PLANFUNCID/RESTORE
NETVIEW://%DYN/EXECUTEPLAN/PLAN/%PLANSTATIONNAME/%PLANFUNCID/IMPLEMENT
NETVIEW://%DYN/EXECUTEPLAN/PLAN/%PLANSTATIONNAME/%PLANFUNCID/REJECT
NETVIEW://%DYN/EXECUTEPLAN/PLAN/%PLANSTATIONNAME/%PLANFUNCID/RESTORE
NETVIEW://%DYN/OMSCOMMAND/ASSIGNCREW/?ID=%SELECTCOMPID&NAME=%SELECTCOMPNAME
NETVIEW://%DYN/OMSCOMMAND/ASSOCIATE/MERGEINCIDENT?ID=%SELECTCOMPID&INCIDENTNAME=
NETVIEW://%DYN/OMSCOMMAND/CREW/DELETE?ID=%ID
NETVIEW://%DYN/OMSCOMMAND/MERGEDOWNSTREAMPREDICTEDINCIDENTS?ID=%SELECTCOMPID
NETVIEW://%DYN/OMSCOMMAND/RESETPREDICTEDSTABLE?ID=%SELECTCOMPID&INCIDENTNAME=
NETVIEW://%DYN/DNAF/RESETLOVFLAG/%NAME
NETVIEW://%DYN/SET/%DEVICETYPENAME/%STATIONNAME/%ID/MANUALDATAENTRY?CONTROLFAILED=FALSE
```